Cooling without Cracking: The Unemployment Cost of Post-Pandemic Labor Market Cooling*

Brendan Epstein† Musa Orak‡ Avani Pradhan§

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Abstract

Post-pandemic labor-market cooling was accompanied by unusually small increases in unemployment. A historically low increase, however, need not be anomalous given the vacancy decline and the initial unemployment rate. Using quarterly data for 17 advanced economies and a common tightness-based episode rule, we compare 14 cooling episodes with peaks in 2022–23 with 35 historical episodes. The median unemployment cost of cooling, the percentage-point increase in unemployment per 100 log points of declining tightness, was 1.16 recently, compared with 2.65 historically and 2.28 outside 2007–08. Every recent episode had a lower cost than its own economy's historical median, and lower starting unemployment accounts for roughly one-quarter of the gap. A vacancy–unemployment relation estimated for the pre-pandemic period, applied to realized vacancy declines and initial unemployment rates, implies a median recent increase of 0.68 percentage point, close to the observed 0.63. Dating episodes from vacancy peaks places the recent response below this benchmark, but neither estimate is precise enough to establish an unusual conditional response. Net participation changes added to recent unemployment rather than holding it down, and inferred exits from unemployment deteriorated less than in historical episodes in quarterly duration measures.

JEL: E24, E32, J63, J64.

Keywords: Beveridge curve; labor-market tightness; vacancies; cooling; soft landing.

*The views in this paper are solely the responsibility of the authors and should not be interpreted as reflecting the views of the Board of Governors of the Federal Reserve System or of any other person associated with the Federal Reserve System. All errors are our own.

†Corresponding author. Department of Economics University of Massachusetts Lowell, Health and Social Sciences Building, Room 347, 113 Wilder St. Lowell, MA 01854. E-mail: epsteinb@gmail.com.

‡Division of International Finance, Board of Governors of the Federal Reserve System, Washington, D.C. 20551, United States. E-mail: musa.orak@frb.gov.

§Division of International Finance, Board of Governors of the Federal Reserve System, Washington, D.C. 20551, United States. E-mail: avani.s.pradhan@frb.gov.

1 Introduction

Unemployment rates across advanced economies rose only modestly from their post-pandemic lows even as job vacancies fell sharply from their peaks. Measured as the ratio of the vacancy rate to the unemployment rate, labor-market tightness declined by about 41 percent in the median economy over the eight quarters following its 2022–23 peak, while unemployment rose by 0.63 percentage point. This combination is central to interpreting a soft landing, yet a given decline in vacancies need not carry the same unemployment increase in labor markets that start from different conditions. A historical comparison can establish how unusual the combination was, but it cannot by itself tell us whether unemployment behaved unexpectedly given the vacancies that disappeared. In this paper, we address both questions by placing post-pandemic cooling in international historical perspective and evaluating it against a vacancy–unemployment relation estimated before the pandemic.

For the historical comparison, we apply one dating rule and a common measure of unemployment cost to all economies and periods. An episode begins at a local peak in log tightness followed by a decline of at least 20 log points over eight quarters. In quarterly data for 17 advanced economies from 1985 to 2026, this rule identifies 35 historical episodes and 14 recent ones. We define an episode's *unemployment cost* as the percentage-point increase in the unemployment rate per 100 log points of declining tightness. This is a descriptive normalization, not a welfare measure or a causal policy tradeoff. For the conditional comparison, we construct a pre-pandemic benchmark: we estimate the elasticity of unemployment to vacancies on pre-2020 eight-quarter windows and apply it to each recent episode's realized vacancy decline and starting unemployment rate. Because tightness contains unemployment, dating on it partly selects episodes on the outcome being assessed, so we repeat both comparisons with episodes dated from vacancy peaks alone.

The historical comparison shows that recent cooling had an unusually low unemployment cost. The median recent cost was 1.16 percentage points per 100 log points of declining tightness, compared with 2.65 in the historical episodes. The financial crisis widens this gap but does not account for it. Episodes beginning in 2007–08 had a median cost of 3.12, and the other historical episodes, at 2.28, still had nearly twice the recent cost. The result is also broad: all 14 recent episodes had lower costs than their own economy's historical median, and the recent median remains the lowest when episodes are dated from vacancy peaks. Part of the gap reflects starting conditions: unemployment began at 4.3 percent in the median recent episode, compared with 5.2 historically, and from a lower base a given proportional

rise in unemployment is mechanically a smaller percentage-point increase. Two accounting exercises assign roughly one-quarter of the gap to this starting point. Most of the remainder reflects a smaller proportional rise in unemployment, which the accounting describes but does not attribute to any particular mechanism.

The pre-pandemic benchmark answers the second question. With an estimated elasticity of 0.37, it implies a median recent increase in the unemployment rate of 0.68 percentage point, compared with 0.63 observed. The median paired residual, the actual increase minus the implied one, is 0.07 point, with a joint bootstrap interval of $[-0.36, 0.24]$. The close fit at the median does not extend to every economy, and it depends on episode dating. When episodes begin at vacancy peaks, the recent median residual is -0.35 point, below the benchmark. The 2007–08 episodes, by contrast, have much larger positive residuals under both dating rules, even when the relation is estimated without the crisis. Because neither recent interval excludes zero, the evidence does not distinguish a close conditional fit from a moderately smaller unemployment response. What it does show is that a relation estimated before the pandemic already associates the realized vacancy declines with small unemployment increases.

Employment and unemployment-duration data show what the modest unemployment increase did and did not conceal. In the 13 recent episodes with complete employment, labor-force, and population data, an exact decomposition attributes +0.38 percentage point of the average unemployment increase to net participation changes and +0.44 point to changes in the employment-to-population ratio.[1] Participation rose on average, so labor-force withdrawal did not hold unemployment down on net. Duration-based flow accounting associates recent cooling with a smaller deterioration in inferred exits from unemployment than in historical episodes, where exits combine moves into employment and out of the labor force. The pattern is clearest in the quarterly duration measures, while longer duration windows give more weight to differences in entry.

The literature presents a range of views on how sharply unemployment must rise when vacancies fall from high levels. Blanchard et al. (2022) emphasize historical U.S. episodes in which vacancies did not fall without a substantial rise in unemployment. In contrast, Figura and Waller (2022, 2024) show how a convex Beveridge curve makes a soft landing plausible when labor tightness starts from historically high levels, while Benigno and Eggertsson (2023)

[1]The stock decomposition requires age-consistent employment, labor-force, and population series. Czechia lacks a 15-and-over employment ratio, and four historical episodes have incomplete data, leaving 13 recent and 31 historical episodes. The flow decomposition requires unemployment-by-duration data, available for all 14 recent episodes and 24 historical ones. Each exercise compares recent and historical episodes within its own sample.

formalize how this Beveridge convexity generates a non-linear Phillips curve where vacancy declines cool price pressures without creating substantial job loss. Bok et al. (2022) reach a similar empirical conclusion for the post-pandemic period by tracing historical paths along the curve and examining when past vacancy contractions coincided with layoffs. Our results suggest that the historical lesson depends on which episodes it draws on: the 2007–08 episodes lie well above the pre-pandemic benchmark, while other historical episodes lie close to it.

A second strand asks what shifts the Beveridge curve, and hence the unemployment associated with a given vacancy rate. Elsby et al. (2015) survey the theory and evidence, while Diamond and Şahin (2015), Hobijn and Şahin (2013), and Hall and Kudlyak (2022) study shifts and persistent outward displacements after earlier downturns. For the pandemic period, Barlevy et al. (2024) study the shifting determinants of the Beveridge curve and Ahn and Rudd (2025) examine the empirical transmission of activity, labor-supply, and reallocation shocks. Proposed mechanisms include variation in recruiting intensity and matching efficiency (Davis et al., 2012; Lubik, 2021) and employer competition to poach already-employed workers (Cheremukhin and Restrepo-Echavarria, 2024). To evaluate whether structural misallocation shifted the frontier outward, studies build on the mismatch index framework of Şahin et al. (2014); applying these metrics to the post-pandemic recovery across advanced economies, Duval et al. (2022) and Pizzinelli and Shibata (2023) find that sectoral and occupational mismatch spiked only briefly before receding, accounting for little of the persistent increase in labor market tightness. Related work uses the Beveridge relation to measure slack and full employment (Crump et al., 2019; Michaillat and Saez, 2021, 2024).

Our flow accounting follows Shimer (2012) and Elsby et al. (2013), who decompose unemployment movements into entry and exit rates for the United States and for OECD economies; together with the stock accounting, it describes the margins of adjustment across economies. Much of the work above studies the United States, and we take the realized vacancy path as given rather than identifying the shocks that produced it. Our contribution is to place the recent adjustment in a consistently dated international record, evaluate it against a relation estimated before the pandemic, and show why the historical and conditional answers need not coincide.

Section 2 describes the panel and the episode definitions, and Section 3 presents the historical comparison. Section 4 evaluates the pre-pandemic benchmark and its sensitivity to episode dating. Section 5 examines employment, participation, and unemployment flows.

Section 6 concludes. The appendices document series construction, the cost-gap accounting, inference, alternative benchmarks, and unemployment flows.

2 Data and episode definitions

2.1 The international panel

Our quarterly panel covers 17 advanced economies from 1985:Q1 to 2026:Q2: Australia, Austria, Belgium, Canada, Czechia, Finland, France, Germany, Japan, Luxembourg, Norway, Portugal, Spain, Sweden, Switzerland, the United Kingdom, and the United States. It contains 1,974 economy-quarters with both vacancies and unemployment, although coverage varies by economy. Vacancies come from OECD series and national sources, including JOLTS for the United States, the ONS Vacancy Survey for the United Kingdom, the CFIB business survey for Canada, and Eurostat job vacancy statistics for Belgium and Spain. Most series measure outstanding vacancies, but France and Japan measure flows of new vacancies. We use seasonally adjusted series where available and retain Canada's series as published.[2] Unemployment rates are harmonized rates for ages 15 and over (16 and over in the United States), and employment comes from labor-force surveys. Following JOLTS, we define the vacancy rate as $v = 100 \times V/(V + E)$, where V denotes vacancies and E employment, and labor-market tightness as $\theta = v/u$, where u is the unemployment rate in percent.

2.2 Episode definitions

We date an episode at a local peak in log tightness, with no higher value within six quarters on either side, followed by a decline of at least 20 log points over the next eight quarters. Each episode's window runs from the peak quarter to eight quarters later. Throughout the paper, log points equal 100 times natural-log differences.

Historical episodes have peaks before 2019:Q1. The rule identifies 35 of them, including 12 with peaks in 2007:Q1–2008:Q4, which we report separately throughout because they differ markedly from the rest. Recent episodes have peaks in 2021:Q3–2023:Q2. There are

[2]Table A1 in Appendix A lists each series, its vintage, and its treatment, including documented breaks and gaps. We chain-link the Czech series across a January 2025 change in vacancy registration, assuming that the entire break is definitional. Section 3 reports comparisons that exclude flow-based vacancy series and affected windows.

14, one in each economy except three: Belgium and Japan, whose declines fall short of the threshold, and Spain, whose tightness rose.

For each episode, let $D = -\Delta_8 \ln \theta > 0$ denote the eight-quarter decline in tightness in natural-log units. Since $\ln \theta = \ln v - \ln u$, the decline splits exactly into a vacancy contribution, $-\Delta_8 \ln v$, and an unemployment contribution, $\Delta_8 \ln u$. We write the unemployment share of the decline as $s = \Delta_8 \ln u / D$. The unemployment cost of an episode is then

$$C = \frac{\Delta_8 u}{D} = u_0 \frac{e^{sD} - 1}{D}, \tag{1}$$

where u_0 is the unemployment rate at the peak. Because D is in natural-log units, C measures the percentage-point increase in unemployment per 100 log points of declining tightness. This normalization makes episodes of different sizes comparable. It also makes the role of starting conditions explicit: holding s and D fixed, the cost is proportional to the starting unemployment rate.

Because tightness contains unemployment, dating on it partly selects episodes on the outcome being assessed: for a given vacancy decline, a larger rise in unemployment makes a window more likely to qualify and can move its peak date. We therefore also date episodes from local peaks in the log vacancy rate followed by a decline of at least 20 log points over eight quarters, with no condition on unemployment or tightness. Section 4 compares the two rules.

3 The unemployment cost in historical perspective

3.1 A broad decline in the cost of cooling

Vacancy-led cooling is not new. Vacancy declines accounted for most of the reduction in tightness in 21 of the 35 historical episodes and in all 14 recent ones (Table 1). Correspondingly, the median unemployment share of the decline is 0.29 recently and 0.39 historically; outside 2007–08, the historical median share is also 0.29. What sets the recent episodes apart is their low unemployment cost. Per 100 log points of declining tightness, the unemployment rate rose 1.16 percentage points in the median recent episode, compared with 2.65 in the median historical episode. The historical median is 3.12 for episodes beginning in 2007–08 and 2.28 for the others. The difference lies in the unemployment increase rather than the size

of the decline: the median tightness declines are similar, 52.6 log points recently and 49.0 historically, but the median unemployment increases are 0.63 and 1.30 percentage points.

Table 1. Cooling episodes, historical and recent

	Historical (peaks 1989–2018)			Recent
	All	2007–08	Other	2022–23
Episodes	35	12	23	14
Medians				
Starting unemployment rate, u_0 (percent)	5.2	4.4	5.4	4.3
Decline in log tightness, D (log points)	49.0	76.4	47.2	52.6
Increase in unemployment rate, $\Delta_8 u$ (pp)	1.30	2.45	1.00	0.63
Unemployment cost, C (pp per 100 log points)	2.65	3.12	2.28	1.16
Unemployment share of the decline, s	0.39	0.53	0.29	0.29
Vacancy-dominated episodes ($s < 0.5$)	21	5	16	14

Notes: An episode is a local peak in log tightness $\ln \theta = \ln(v/u)$, with no higher value within six quarters on either side, followed by a decline of at least 20 log points over the next eight quarters. All changes are measured from the peak quarter to eight quarters later. Historical episodes have peaks before 2019:Q1; the 2007–08 column covers peaks in 2007:Q1–2008:Q4. Recent episodes have peaks in 2021:Q3–2023:Q2. The cost C and share $s = \Delta_8 \ln u / D$ are defined in equation (1); vacancy-dominated episodes are those in which vacancies account for more than half of the decline. Medians are taken across episodes. Table A2 reports the historical statistics for stock-vacancy economies, for windows that avoid documented breaks, and for windows ending by 2019:Q4. Source: OECD, Eurostat, national sources, and authors' calculations.

Figure 1 shows that the result holds without forming ratios and within economies. In Panel A, all 14 recent episodes lie below the descriptive line fitted to the 35 historical episodes, whose slope is 2.1 percentage points per 100 log points. Panel B compares each recent episode with its own economy's historical median, which holds country composition fixed. The recent cost is lower in all 14 economies, by 1.19 points in the median economy (Table A7). Excluding 2007–08 from each economy's history leaves 11 comparisons, of which 9 have a lower recent cost (Figure A4).

3.2 Starting unemployment and the cost gap

Lower starting unemployment accounts for part, but not all, of the cost gap. The median recent episode began at 4.3 percent unemployment, compared with 5.2 percent historically. By equation (1), holding s and D fixed, a lower starting rate implies a proportionally lower cost. Replacing each recent starting rate with the historical median closes 32 percent of the

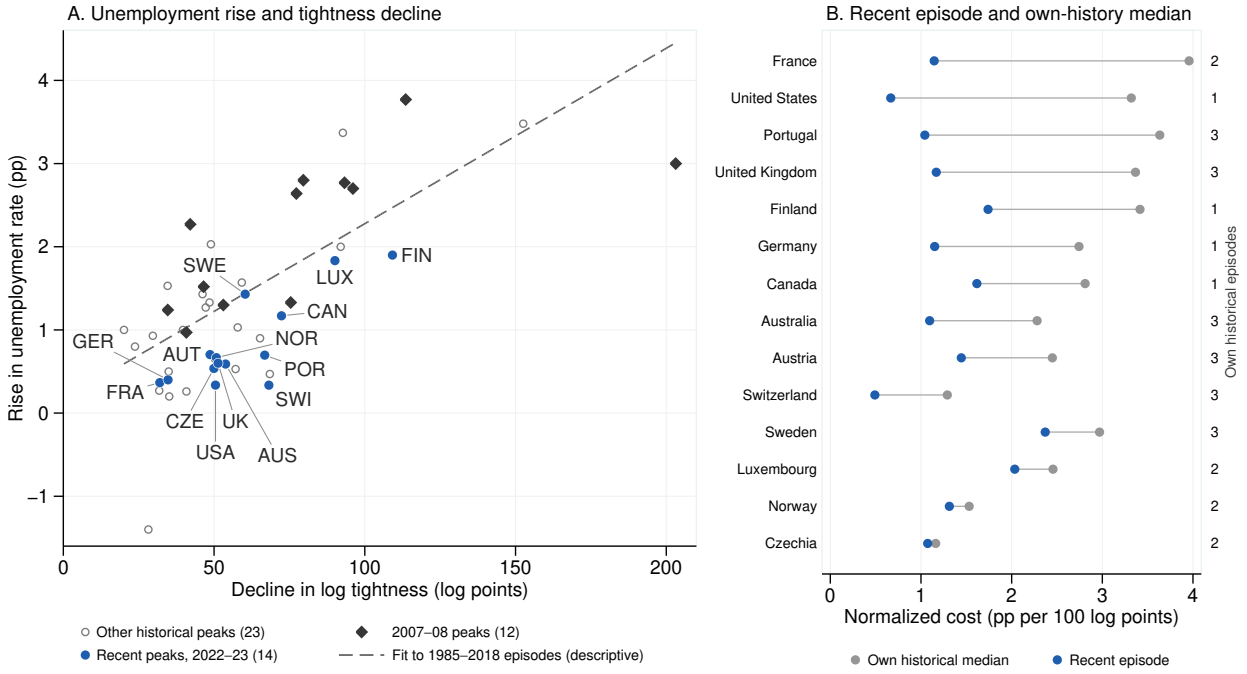


Figure 1. The unemployment cost of cooling across episodes and economies. Panel A plots the eight-quarter increase in the unemployment rate against the decline in log tightness for the 35 historical episodes (gray circles; 2007–08 peaks as dark diamonds) and the 14 recent episodes (blue, labeled). The dashed line is a descriptive least-squares fit to the historical episodes, not a tradeoff. Panel B compares each recent cost (blue) with the median cost of the same economy's historical episodes (gray), in percentage points of unemployment per 100 log points of declining tightness. Economies are sorted by the paired difference, largest reduction at the top; the number of historical episodes appears at the right. Source: OECD, Eurostat, national sources, and authors' calculations.

1.49-point gap in median cost. Averaging over the order in which starting unemployment and the unemployment share are replaced assigns 28 percent to starting unemployment, and an exact decomposition of the difference in mean log cost assigns 23 percent. These two allocations put the contribution of starting unemployment at roughly one-quarter of the gap, while weighting economies equally rather than episodes raises the order-averaged share to 40 percent. Most of the remaining gap reflects the smaller recent unemployment share, which accounts for 47 and 69 percent in the two allocations. Outside 2007–08, where the median shares are equal, starting unemployment accounts for 39 percent of the smaller 1.12-point gap and the unemployment share for only 8 percent (Appendix B).

3.3 Robustness and inference

The historical comparison does not depend on the least comparable data. Excluding France and Japan, whose vacancy series measure flows rather than stocks, gives a historical median cost of 2.70. Excluding historical windows that touch a documented break leaves it at 2.65, and keeping only windows that end by 2019:Q4 gives 2.52 (Table A2). Each remains well above the recent median cost of 1.16.

Inference must allow for two forms of dependence: economies contribute several episodes, and peaks cluster in global waves. The recent episodes, in particular, form a single wave. Resampling economies gives a 95 percent interval of $[-1.87, -0.88]$ for the recent-minus-historical difference in median cost, and resampling five-year calendar blocks of historical peaks gives $[-1.89, -0.62]$. Both intervals exclude zero. Inference across global waves is necessarily coarser. Grouping peaks within six quarters of one another yields 7 historical waves, with median costs from 1.4 to 4.1, and the recent median lies below all of them. With seven historical waves and one recent wave, however, even this lowest possible rank has a one-sided probability of 1/8 under exchangeability (Appendix C).

4 A pre-pandemic benchmark

4.1 Design

The historical comparison shows that recent cooling had a low unemployment cost. It cannot say whether that response was unusual given the vacancies that disappeared and the unemployment rate from which cooling began. The pre-pandemic benchmark addresses this

question. It asks how large an unemployment increase a vacancy–unemployment relation estimated before the pandemic would associate with each economy's realized vacancy decline and starting unemployment rate. On all overlapping eight-quarter windows that end by 2019:Q4 and do not touch a documented break, we estimate

$$\Delta_8 \ln u_{it} \;=\; \alpha_i \;-\; \beta\, \Delta_8 \ln v_{it} \;+\; \varepsilon_{it}, \tag{2}$$

where $\Delta_8 \ln u_{it}$ and $\Delta_8 \ln v_{it}$ are the changes in the log unemployment and vacancy rates of economy i from quarter t to $t+8$, α_i is an economy fixed effect, and β is the elasticity of unemployment to vacancies. The minus sign makes β positive when unemployment rises as vacancies fall. The estimate is $\hat{\beta} = 0.37$ on 1,309 windows, with a standard error of 0.04 clustered by economy and 0.02 using the method of Driscoll and Kraay (1998). Leaving out one economy at a time gives estimates between 0.35 and 0.39, and the four- and twelve-quarter versions give 0.33 and 0.40.

For an episode beginning at unemployment rate u_{0i} with realized vacancy change $\Delta_8 \ln v_i$, in natural-log units, the benchmark unemployment increase is

$$\widehat{\Delta_8 u_i} \;=\; u_{0i} \Big[\exp\Big(-\hat{\beta}\, \Delta_8 \ln v_i \Big) - 1 \Big]. \tag{3}$$

We use the common slope and set the fixed effect to zero, so the calculation isolates the unemployment change associated with vacancies from each economy's average unemployment drift; adding the estimated fixed effects back is a sensitivity exercise. The residual is the episode's actual unemployment change minus its implied change, computed for each episode before taking medians. All recent windows lie outside the estimation sample. Because the benchmark conditions on the realized vacancy path, it is a conditional historical comparison rather than a forecast, and it identifies no shock.

4.2 Results

Under tightness dating, the benchmark closely matches the median recent unemployment increase. Figure 2 and Table 2 report the episode results, and Table A9 in Appendix D compares specifications. Given a median vacancy decline of 40.0 log points, equation (3) implies a median increase of 0.68 percentage point, compared with 0.63 observed. The median paired residual is 0.07 point, and actual unemployment rose less than implied in 5 of 14

economies. A bootstrap that resamples both the economies used to estimate $\hat{\beta}$ and the recent episodes gives a 95 percent interval of $[-0.36, 0.24]$ for the median residual.

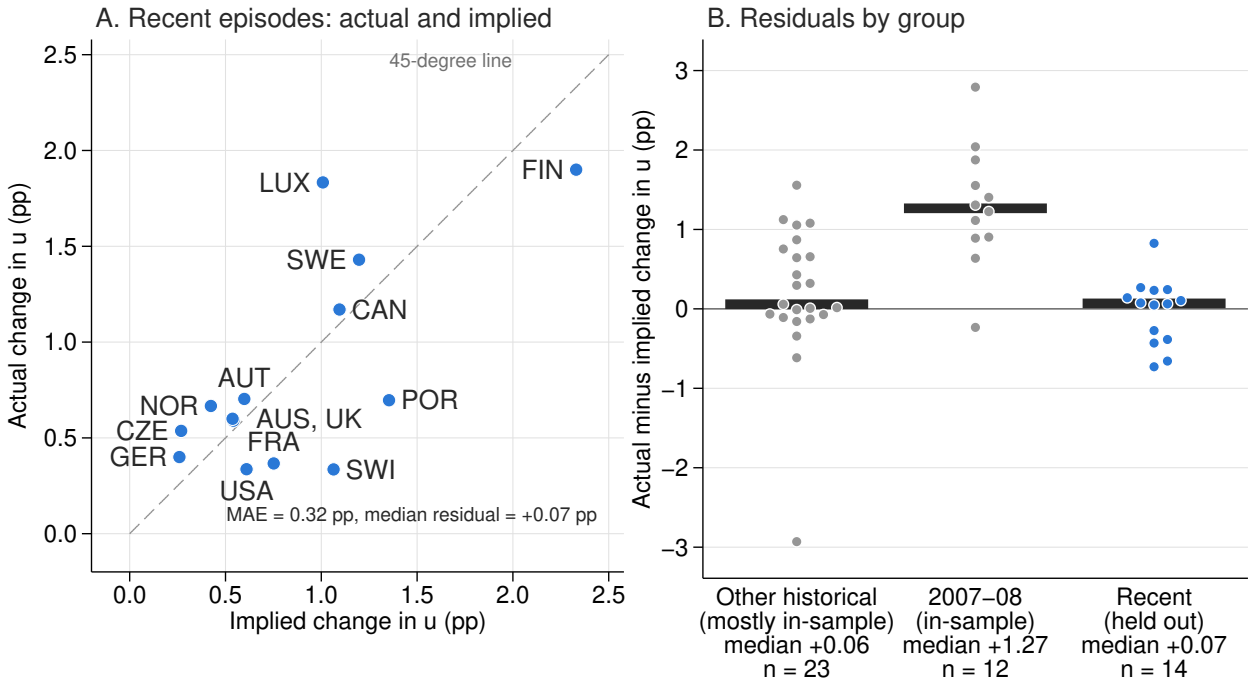


Figure 2. Actual unemployment changes and the pre-pandemic benchmark. Panel A plots the actual eight-quarter increase in the unemployment rate after each economy's 2022–23 tightness peak against the increase implied by equation (3); the dashed line is the 45-degree line. The elasticity in equation (2) is estimated on windows ending by 2019:Q4 and applied to each episode's realized vacancy decline and starting unemployment rate. Panel B shows residuals, actual minus implied, for the historical episodes outside 2007–08, the 2007–08 episodes, and the recent episodes; bars denote group medians. Of the 35 historical windows, 29 enter estimation, including 10 of the 12 with peaks in 2007–08. All recent windows lie outside the estimation sample. Source: authors' calculations.

Agreement at the median is not equivalence. The benchmark captures much of the variation across economies, with a correlation of 0.70 between actual and implied increases, but its errors are material. The mean absolute residual is 0.32 percentage point, and individual residuals range from −0.73 in Switzerland to 0.83 in Luxembourg (Figure 2, Panel A). Adding each economy's fixed effect leaves the fit about unchanged. Economy-specific slopes preserve the close median comparison but roughly double the root mean squared error, to 0.83 point. At four- and twelve-quarter horizons, the median residuals are 0.00 and 0.11 percentage point (Table A11), so the result does not rest on the eight-quarter window.

Table 2. Benchmark residuals under two episode definitions and two estimation samples

			Median residual (pp)			Recent: joint
Estimation sample	Windows	$\hat{\beta}$	Other historical	2007–08	Recent	95% interval
A. Tightness dating						
All pre-2020	1,309	0.37	0.06	1.27	0.07	[−0.36, 0.24]
Excluding 2007–09	1,053	0.34	0.26	1.40	0.14	[−0.24, 0.31]
B. Vacancy-peak dating						
All pre-2020	1,309	0.37	−0.13	0.45	−0.35	[−0.63, 0.18]
Excluding 2007–09	1,053	0.34	0.01	0.58	−0.21	[−0.50, 0.22]

Notes: Residuals are actual minus implied eight-quarter increases in the unemployment rate, in percentage points, with implied increases from equation (3) and the fixed effect set to zero. Residuals are computed for each episode before taking medians, so a median residual is not the difference between median actual and median implied increases. Tightness dating uses the episode rule of Section 2. Vacancy-peak dating requires a local peak in $\ln v$ followed by an eight-quarter decline of at least 20 log points, with no condition on tightness or unemployment. Historical peaks span 1989–2018 under tightness dating and 1987–2018 under vacancy-peak dating; 2007–08 denotes peaks in 2007:Q1–2008:Q4, and recent episodes have peaks in 2021:Q3–2023:Q2. Episode counts (other historical, 2007–08, recent) are 23, 12, and 14 under tightness dating and 25, 10, and 14 under vacancy-peak dating; one further other historical vacancy episode (Austria 1990:Q2) lacks unemployment data. Equation (2) is estimated on eight-quarter windows in the 17 economies that end by 2019:Q4 and do not touch a documented break. The second sample also drops every window touching 2007:Q1–2009:Q4; this is an event holdout rather than a real-time forecast, since later windows remain in the sample. Under tightness dating, 10 of the 12 windows with peaks in 2007–08 and 19 of the 23 other historical windows enter the pre-2020 sample; no 2007–08 window and 19 other historical windows enter the holdout sample. Recent windows lie outside both samples. Joint intervals are the 2.5th and 97.5th percentiles of 999 economy-cluster bootstrap replications that resample the 17 economies, re-estimate $\hat{\beta}$, and resample recent episodes with their economies. Source: authors' calculations.

The 2007–08 episodes have much larger positive residuals than either the recent episodes or the other historical episodes. In the median 2007–08 episode, unemployment rose 1.27 percentage points more than the benchmark implies. In the median historical episode outside 2007–08, it rose 0.06 point more, close to the recent value of 0.07. When each historical residual is recomputed from the benchmark estimated without that episode's economy, the recent median residual lies at percentile 52 of the other historical residuals and percentile 8 of the 2007–08 residuals. A historical reference drawn heavily from the financial crisis therefore gives a different impression of the recent adjustment.

The contrast persists when the crisis is excluded from estimation. This check matters because most historical windows enter the estimation sample, and leaving out one economy does not remove a global crisis from the other sixteen. Dropping every window that touches

2007:Q1–2009:Q4 gives $\hat{\beta} = 0.34$ on 1,053 windows. That relation implies a median recent increase of 0.61 percentage point, with a median residual of 0.14 and a joint interval of $[-0.24, 0.31]$. The 2007–08 median residual rises to 1.40 points, while the other historical median is 0.26 (Table 2, Panel A; Figure A5). The benchmark thus delivers a small recent increase without the crisis windows, and the crisis episodes stand out against a relation that never saw them. Because later historical windows remain in the sample, this is an event holdout rather than a real-time forecast evaluation.

Relative to each economy's pre-2020 Beveridge curve, the median recent episode began 51.8 log points above the curve and ended 19.3 above it, whereas the median historical episode began 2.5 above and ended 16.7 below. These distances locate observations relative to fitted curves rather than measuring changes in matching efficiency (Appendix D).

4.3 Vacancy-peak dating

Vacancy-peak dating places the recent unemployment response below the benchmark while preserving the contrast with 2007–08. The vacancy rule, defined in Section 2, selects 26 other historical episodes, 10 in 2007–08, and 14 recent episodes in the same economies as the tightness rule; one other historical episode lacks an unemployment outcome. Median unemployment increases are 0.57, 1.65, and 0.47 percentage points, respectively, and median residuals are -0.13, 0.45, and -0.35 (Table 2, Panel B). The recent median residual has a joint interval of $[-0.63, 0.18]$, and actual unemployment rose less than implied in 8 of 14 economies. The ordering survives excluding the crisis from estimation and raising the threshold to 30 log points, which gives median residuals of -0.01, 0.60, and -0.43 (Table A13). The magnitudes differ, however: the 2007–08 excess is about a third of its size under tightness dating, and the recent median moves from just above the benchmark to below it.

The difference between the recent estimates comes from timing. Seven of the fourteen economies keep the same peak quarter; six vacancy peaks come one quarter earlier and one comes one quarter later (Figure A6). These shifts change the starting unemployment rate, the realized vacancy decline, and the endpoint at which unemployment is measured. Selecting on tightness thus enlarges the 2007–08 excess and dates the recent windows so that the median response matches the benchmark rather than falling below it. It does not create the contrast.

Because neither recent interval excludes zero, the two rules do not distinguish a close conditional fit from a moderately smaller unemployment response. They do show that the low recent unemployment cost extends to a rule based on vacancies alone. Among vacancy-dated

episodes in which tightness fell, the median cost is 1.15 recently, compared with 1.62 in other historical episodes and 2.55 in 2007–08.

4.4 Interpretation

Together, the historical comparison and the pre-pandemic benchmark give starting conditions and episode dating a concrete role in interpreting the soft landing. Low starting unemployment accounts for part of the historical cost gap, and the 2007–08 episodes lie well above the benchmark under both dating rules and both estimation samples. Recent costs nonetheless remain low relative to historical episodes outside 2007–08, 1.16 against 2.28, and neither starting unemployment nor any single channel accounts for the full difference. Because the benchmark takes the realized vacancy path as given, it also cannot say whether the same starting labor market would have absorbed a recessionary shock as easily. Section 5 examines the margins through which labor markets adjusted.

5 Margins of labor-market adjustment

5.1 Employment and participation

Net participation changes added to recent unemployment rather than concealing an employment decline. Let E, L, and P denote employment, the labor force, and the age-consistent population. Since $1 - u/100 = (E/P)/(L/P)$, the eight-quarter change in the unemployment rate decomposes exactly into an employment-to-population contribution and a participation contribution. A fall in E/P raises unemployment, as does a rise in L/P, holding the other ratio fixed. A small reconciliation term links this stock-based change to the harmonized unemployment series (see the notes to Table 3). In the 13 recent episodes with complete data, participation contributed +0.38 percentage point to the average unemployment increase, and employment contributed +0.44 point (Table 3). In the 20 historical episodes outside 2007–08, the corresponding contributions were +0.38 and +0.86 point. Participation therefore raised recent unemployment on average rather than holding it down. The recent employment contribution is smaller, although the difference is imprecisely estimated and sensitive to age coverage. The 2007–08 episodes show what labor-force withdrawal looks like in these data: participation contributed -0.80 point, offsetting part of an employment contribution of +2.86 points. Net stock changes can still conceal offsetting gross flows, which we examine next.

Table 3. Employment and participation contributions to eight-quarter unemployment changes

			Contribution of		
Episode group	N	$\Delta_8 u$	Employment, C_E	Participation, C_{LFP}	Reconciliation
Historical					
All	31	+1.56	+1.57	−0.04	+0.03
2007–08	11	+2.12	+2.86	−0.80	+0.06
Other	20	+1.25	+0.86	+0.38	+0.01
Recent (2022–23)	13	+0.85	+0.44	+0.38	+0.03

Notes: Group means of episode-level contributions, in percentage points. Employment, participation, and reconciliation sum to the change in the harmonized unemployment rate, $\Delta_8 u$. The employment and participation contributions decompose the change in the stock unemployment rate exactly, $\Delta u^{EL} = C_E + C_{LFP}$; reconciliation equals $\Delta_8 u - \Delta u^{EL}$. A positive employment contribution reflects a fall in employment relative to population, and a positive participation contribution reflects a rise in participation. The sample uses ages 15 and over (16 and over in the United States). Czechia lacks the required employment ratio and four historical episodes have incomplete data, so the sample is smaller than the main episode sample. Source: OECD, Eurostat, national sources, and authors' calculations.

5.2 Entry into and exit from unemployment

Duration-based flow accounting points to a smaller deterioration in unemployment exits during recent cooling. Following Shimer (2012) and Elsby et al. (2013), we infer entry and exit hazards from unemployment-duration data, available for 24 historical and all 14 recent episodes (Appendix E). An exact dynamic decomposition assigns a median 7.0 log points of the recent unemployment increase to reduced exits, compared with 22.3 historically, while median entry contributions are about 16 log points in both groups (Table A15). Conditional on the vacancy decline, recent exit contributions lie a median 10.1 log points below the historical relation, with negative residuals in 11 of 14 episodes. The pattern holds with four-quarter averages and a six-month duration window. Longer windows shift weight toward entries: with a twelve-month window, both recent contributions are lower than historically, and annual OECD data locate the difference mainly in entries (Table A16).

Because the inferred exit hazard includes moves out of the labor force as well as into jobs, net participation growth can coexist with gross exits into inactivity offset by other entries. The stock and flow results therefore describe complementary aspects of the adjustment. Together, they associate recent cooling with net participation that added to unemployment

rather than holding it down and, in the quarterly duration measures, with less deterioration in unemployment exits.

6 Conclusion

Post-pandemic labor-market cooling had an unusually low unemployment cost by historical standards, but the unemployment increase was not demonstrably unusual relative to a pre-pandemic benchmark. The median recent cost was 1.16 percentage points per 100 log points of declining tightness, compared with 2.65 across historical episodes and 2.28 outside 2007–08. Every recent episode had a lower cost than its own economy's historical median, and lower starting unemployment accounts for roughly one-quarter of the gap. Applied to each economy's realized vacancy decline and starting unemployment rate, a relation estimated before 2020 implies a median increase of 0.68 percentage point, close to the observed 0.63 under tightness dating. Under vacancy-peak dating, the observed increase falls below the benchmark. Neither residual is precise enough to distinguish a close conditional fit from a moderately smaller unemployment response.

Two choices shape how the recent episodes compare with history: which historical episodes serve as the reference, and how episodes are dated. The 2007–08 episodes lie well above the benchmark, with a median residual of 1.27 percentage points against 0.06 in other historical episodes, and they remain distinctive when the relation is estimated without the crisis windows. A reference drawn heavily from the financial crisis therefore implies a larger unemployment counterpart to vacancy declines than the rest of the historical record does. Episode dating matters as well. Because tightness contains unemployment, dating on it partly selects episodes on the outcome being assessed; dating from vacancy peaks moves half of the recent windows by one quarter and shifts the recent median residual from just above the benchmark to below it. Neither choice overturns the low recent cost, which remains below other historical episodes outside 2007–08 and under vacancy-peak dating. The two comparisons respond differently because they answer different questions, and they should not be read as competing estimates of one structural tradeoff.

The stock and flow evidence shows which margins absorbed the adjustment. Net participation changes added to recent unemployment on average rather than holding it down, whereas in 2007–08 participation fell and offset part of a much larger employment contribution. Inferred exits from unemployment also deteriorated less than in historical episodes, a pattern

clearest in the quarterly duration measures. Because these exits include moves into inactivity as well as into jobs, the two results are complementary: net participation growth can coexist with gross exits from the labor force offset by other entries.

Three features of the evidence bound these conclusions. National vacancy concepts differ, historical episodes cluster in a small number of global waves, and the benchmark takes the realized vacancy path as given, so it is neither a forecast of the next soft landing nor an evaluation of the policies behind this one. Within those limits, the paper provides a consistently dated international record of cooling and an explicit pre-pandemic benchmark for interpreting it. Identifying the shocks and gross worker transitions behind the recent adjustment remains a question for future work.

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A Data sources

Table A1 lists the vacancy concept, seasonal adjustment, coverage, vintage, and treatment for each economy. Three links connect series across source changes. For the United Kingdom, OECD coverage ends in 2023:Q2 and the ONS Vacancy Survey (series AP2Y, three-month averages) continues it; ONS levels are multiplied by 1.010, the ratio of the two series in the four quarters before the join. For Japan, the Ministry of Health, Labour and Welfare's seasonally adjusted new-job-openings series is used from 2012:Q3, with a link factor of 1.000. For Czechia, the vacancy register changed in January 2025 (Czech Statistical Office, 2025), and the measured level fell by about 100 log points. The OECD flags the break but publishes no reconstruction, so we multiply earlier levels by 0.36 to make the series continuous. This chain-link assumes that the entire break is definitional. Czechia's eight-quarter window ends in 2024:Q1, before the break; the adjustment affects only the twelve-quarter comparison in Table A11.

Unemployment rates are harmonized rates for ages 15 and over (16 and over in the United States), with monthly observations averaged to quarters where available. Employment comes from labor-force surveys through 2025:Q3 and is extended thereafter using the labor force times one minus the unemployment rate. Values from 2024:Q1 onward are the September 2026 vintage; earlier values come from an earlier download of the same series. Within the overlap, the two vintages differ by at most 1.3 percent for vacancies, apart from one U.S. quarter in 2001, and by at most 0.53 percentage point for unemployment rates.

Several historical and later-series breaks constrain which episode windows can be compared without adjustment. The vacancy series contain documented breaks and gaps: a 2008–09 gap in Australia, a 1999 break in Germany, a 2001 step in the United Kingdom, a 2013 break in Spain, the 2025 Czech registration change, the 2012 source change in Japan, and quarters before 2010 in which Swiss unemployment is observed only in the second quarter. Every eight-quarter window that touches one of these quarters is flagged; no recent window is affected, and 31 of the 35 historical episodes remain when the flagged windows are dropped. Tightness as defined here, the ratio of the vacancy rate to the unemployment rate, differs from vacancies per unemployed worker by the factor $(E + U)/(E + V)$, which averages 5.6 log points in the panel and whose eight-quarter changes have a standard deviation of 1.9 log points.

Table A1. Data sources by economy

Economy	Vacancy series	Concept	Seasonal adjustment	Coverage	Vintage	Treatment
Australia	OECD DF_OIALAB_INDIC AUS.VAC_U.PS._Z.Y.S1.Q (ABS quarterly survey)	Stock: job vacancies	By source	1985:Q1–2026:Q1 (gap 2008:Q3–2009:Q3)	Standard	None (source break 2008:Q2; 2008:Q3–2009:Q3 missing in source)
Austria	OECD DF_OIALAB_INDIC AUT.VAC_U.PS._Z.Y.S1.M, mean of months	Stock: unfilled vacancies	By source	1993:Q1–2026:Q2	Standard	None
Belgium	OECD BEL.VAC_U.PS._Z.Y.S1.M to 2004:Q1; Eurostat jvs_q_nace2 SA B–S from 2013:Q1	Stock: unfilled vacancies (Eurostat JVS from 2013)	By source	1985:Q1–2026:Q2 (gaps 1998:Q1–1998:Q4, 2004:Q2–2012:Q4)	Standard; vacancies from 2025:Q4 in latest release	Ratio-linked 2025:Q3; growth-linked 2025:Q4 to NACE Rev. 2.1 (jvs_q_r21); gap unlinked
Canada	CFIB Help Wanted: number of vacant positions, Canada (quarterly data file)	Stock: private-sector vacant positions (CFIB survey)	Not stated by CFIB; used as published	2004:Q1–2026:Q2	Standard; vacancies from 2025:Q4 in CFIB 2026Q2 release	Linked at 2025:Q3 to the latest CFIB release (2026Q2); CFIB revises history
Czechia	OECD DF_OIALAB_INDIC CZE.VAC_U.PS._Z.Y.S1.M, mean of months	Stock: registered unfilled vacancies	By source	1998:Q1–2026:Q1	Standard	Chain-linked across the January 2025 registration change (factor 0.36); an assumption, not an official reconstruction
Finland	OECD DF_OIALAB_INDIC FIN.VAC_U.PS._Z.Y.S1.M, mean of months	Stock: unfilled vacancies	By source	1998:Q1–2026:Q2	Standard	None
France	OECD DF_OIALAB_INDIC FRA.VAC_N.PS._Z.Y.S1.M, mean of months	Flow: new vacancies registered in the period	By source	2003:Q1–2026:Q2	Standard	None
Germany	OECD DF_OIALAB_INDIC DEU.VAC_U.PS._Z.Y.S1.M, mean of months	Stock: unfilled vacancies	By source	1991:Q1–2026:Q2	Standard	None (OECD break flag November 1999 not adjusted)
Japan	OECD JPN VAC_N to 2012:Q2; MHLW new job openings (e-Stat) from 2012:Q3	Flow: new job openings (excl. new graduates, incl. part-time)	By source	1985:Q1–2026:Q2	Standard; vacancies from 2025:Q4 in latest release	MHLW series linked at 2012:Q2 (factor 1.000)
Luxembourg	OECD DF_OIALAB_INDIC LUX.VAC_U.PS._Z.Y.S1.M, mean of months	Stock: unfilled vacancies	By source	2007:Q1–2026:Q1	Standard	None
Norway	OECD DF_OIALAB_INDIC NOR.VAC_U.PS._Z.Y.S1.M, mean of months	Stock: unfilled vacancies	By source	2000:Q1–2026:Q2	Standard	None
Portugal	OECD DF_OIALAB_INDIC PRT.VAC_U.PS._Z.Y.S1.M, mean of months	Stock: unfilled vacancies	By source	1998:Q1–2026:Q2	Standard	None

continued on the next page

Table A1, continued

Economy	Vacancy series	Concept	Seasonal adjustment	Coverage	Vintage	Treatment
Spain	OECD ESP.VAC_U.PS._Z.Y.S1.M to 2005:Q1; Eurostat jvs_q_nace2 SA B–S from 2010:Q1	Stock: unfilled vacancies (Eurostat JVS from 2010)	By source	1999:Q1–2026:Q2 (gap 2005:Q2–2009:Q4)	Standard; vacancies from 2025:Q4 in latest release	Ratio-linked 2025:Q3; growth-linked 2025:Q4 to NACE Rev. 2.1 (jvs_q_r21); gap unlinked
Sweden	OECD DF_OIALAB_INDIC SWE.VAC_U.PS._Z.Y.S1.M, mean of months	Stock: unfilled vacancies	By source	2001:Q1–2026:Q2	Standard	None
Switzerland	OECD DF_OIALAB_INDIC CHE.VAC_U.PS._Z.Y.S1.M, mean of months	Stock: unfilled vacancies	By source	1999:Q2–2025:Q4 (Q2 only in 1999–2009)	Standard	None (seasonally adjusted series ends 2025:Q4)
United Kingdom	OECD GBR VAC_U to 2023:Q2 (ONS Vacancy Survey); ONS AP2Y from 2023:Q3	Stock: unfilled vacancies	By source	1985:Q1–2026:Q2	Sept. 2026 download from 2024:Q1	ONS series linked at 2023:Q2 (factor 1.010); untreated 2001 step
United States	FRED JTSJOL (JOLTS job openings), mean of months	Stock: job openings (JOLTS)	By source	2000:Q4–2026:Q2	Standard	None (2000:Q4 is December 2000 only)

Notes: Unemployment rates are harmonized and seasonally adjusted, for ages 15 and over (16 and over in the United States), with monthly values averaged to quarters (quarterly series for Switzerland and the United Kingdom). Unemployment rates and the labor force come from the OECD dataflows *Monthly unemployment rates* and *Labour force population*, and from FRED for the United States. Coverage is the span with both vacancies and unemployment. "Standard" vintage means values from 2024:Q1 onward come from a September 2026 download and earlier values from an earlier download of the same series; within the joint sample, the two differ by at most 1.3 percent for vacancies (2.3 percent for the United States in 2001) and 0.53 percentage point for unemployment rates.

Table A2. Historical episodes: alternative samples

	All	Stock-vacancy economies	No break in window	Window ends by 2019:Q4
Episodes	35	30	31	33
Medians				
Starting unemployment rate, u_0 (percent)	5.2	5.2	5.4	5.0
Decline in log tightness, D (log points)	49.0	51.7	50.4	50.4
Increase in unemployment rate, $\Delta_8 u$ (pp)	1.30	1.32	1.33	1.33
Unemployment cost, C (pp per 100 log points)	2.65	2.70	2.65	2.52
Unemployment share of the decline, s	0.39	0.39	0.39	0.39
Vacancy-dominated episodes ($s < 0.5$)	21	18	19	21

Notes: Historical episodes are defined as in Table 1, with peaks in 1989–2018. Stock-vacancy economies exclude France and Japan, whose vacancy series measure flows of new vacancies. The no-break sample drops episodes whose eight-quarter window touches a documented break or gap (Table A1 and Appendix A). The 2019:Q4 sample drops the two episodes with peaks in 2018 (Sweden 2018:Q1 and Luxembourg 2018:Q4), whose windows end in 2020. Vacancy-dominated episodes are those in which vacancies account for more than half of the decline in tightness. Source: OECD, Eurostat, national sources, and authors' calculations.

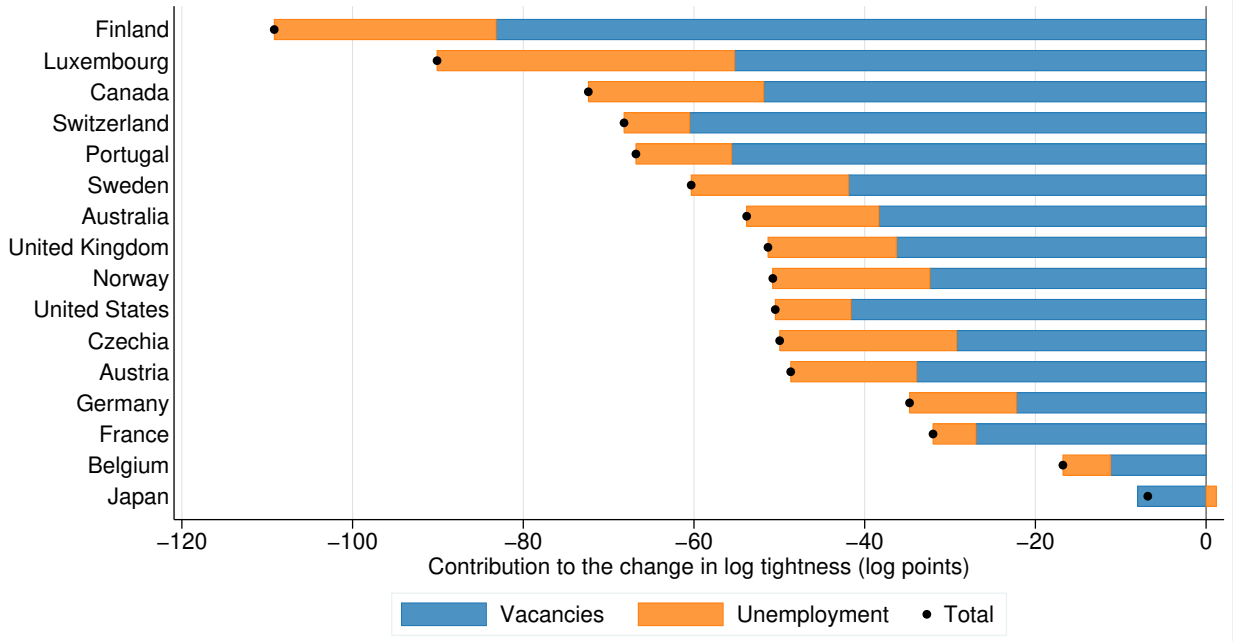


Figure A1. Vacancy and unemployment contributions to the change in log tightness. Contributions of vacancies (blue) and unemployment (orange) to the change in log tightness over the eight quarters after each economy's 2022–23 tightness peak; black dots mark the total; Spain, whose tightness rose, is omitted. Source: OECD, Eurostat, national sources, and authors' calculations.

Table A3. The 2022–23 cooling episodes, eight quarters after each economy's tightness peak

			Log points			Vacancy		
Economy	Peak	u_0	D	$-100\Delta_8 \ln v$	$100\Delta_8 \ln u$	share	$\Delta_8 u$	C
A. Recent episodes								
Finland	2022:Q2	6.4	109.2	83.2	26.0	0.76	1.90	1.74
Luxembourg	2022:Q2	4.4	90.1	55.3	34.8	0.61	1.83	2.03
Canada	2022:Q2	5.1	72.4	51.8	20.5	0.72	1.17	1.62
Switzerland	2022:Q3	4.2	68.2	60.5	7.7	0.89	0.34	0.49
Portugal	2022:Q1	5.9	66.8	55.6	11.2	0.83	0.70	1.04
Sweden	2022:Q3	7.1	60.3	41.9	18.4	0.69	1.43	2.37
Australia	2022:Q3	3.5	53.8	38.3	15.5	0.71	0.59	1.10
United Kingdom	2022:Q1	3.7	51.3	36.3	15.0	0.71	0.60	1.17
Norway	2022:Q1	3.3	50.8	32.4	18.4	0.64	0.67	1.31
United States	2022:Q2	3.6	50.5	41.6	8.9	0.82	0.34	0.67
Czechia	2022:Q1	2.3	50.0	29.2	20.7	0.59	0.54	1.07
Austria	2022:Q2	4.4	48.7	33.9	14.7	0.70	0.70	1.45
Germany	2022:Q2	3.0	34.7	22.2	12.5	0.64	0.40	1.15
France	2023:Q1	7.1	32.0	27.0	5.0	0.84	0.37	1.15
Median, 14 episodes		4.3	52.6	40.0	15.3	0.71	0.63	1.16
B. Economies without a recent episode								
Belgium	2022:Q1	5.3	16.8	11.2	5.5	0.67	0.30	1.81
Japan	2022:Q4	2.5	6.8	8.0	−1.2	1.17	−0.03	−0.44
Spain	2023:Q3	11.9	−7.2	5.6	−12.8	n.a.	−1.43	n.a.
Median, 16 economies with declining tightness		4.3	51.0	37.3	14.9	0.71	0.59	1.16

Notes: Changes from each economy's tightness peak to eight quarters later. Panel A lists the 14 recent episodes, with peaks in 2021:Q3–2023:Q2, sorted by the size of the decline in tightness. Panel B lists the three economies without a recent episode: the declines in Belgium and Japan fall short of the 20-log-point threshold, and Spain's tightness rose; for these economies the peak is the highest log tightness in 2021:Q1–2023:Q4. u_0 is the unemployment rate at the peak (percent); $D = -100\Delta_8 \ln \theta$ is the decline in log tightness, which splits into the vacancy contribution $-100\Delta_8 \ln v$ and the unemployment contribution $100\Delta_8 \ln u$. The vacancy share is the vacancy contribution divided by D, equal to $1 - s$. $\Delta_8 u$ is the change in the unemployment rate (percentage points), and C is the unemployment cost from equation (1) (percentage points per 100 log points). The share and cost are not defined for Spain. The 16-economy median excludes Spain. Source: OECD, Eurostat, national sources, and authors' calculations.

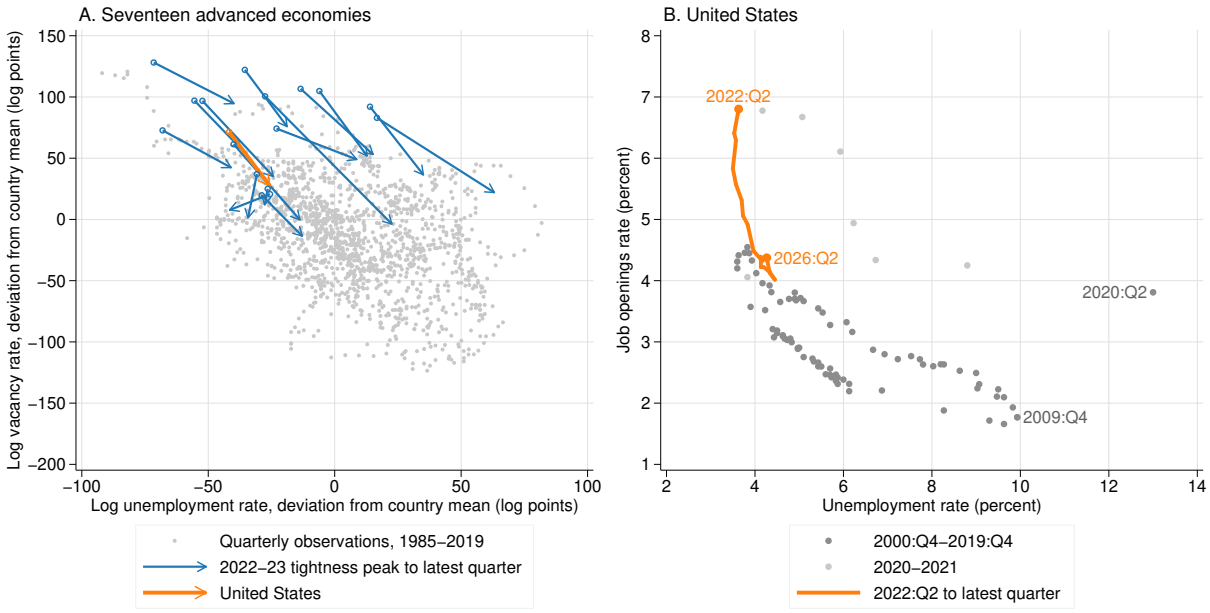


Figure A2. Beveridge curves. Panel A: log vacancy rate against log unemployment rate, each as a deviation from the economy's own mean, for the 17 economies; gray dots are quarterly observations from 1985 to 2019 and each arrow runs from the economy's 2022–23 tightness peak to its latest quarter, with the United States in orange. Panel B: the United States job openings rate against the unemployment rate; dark gray dots are 2000:Q4 to 2019:Q4, light gray dots 2020–21, and the orange line 2022:Q2 to 2026:Q2. Source: OECD, Bureau of Labor Statistics, and authors' calculations.

B Accounting for the cost gap

Equation (1) separates starting unemployment from the unemployment share and the size of the decline, without identifying either as a causal mechanism. It writes the cost as the product of the starting rate and a function of the unemployment share s, which this appendix and Table A4 also call the proportional response, and the decline D. The recent episodes began from a lower rate, 4.3 percent in the median episode against 5.2 historically, and with a smaller unemployment share, 0.29 against 0.39; their declines were of similar size, 52.6 log points against 49.0. Figure A3 shows the full distributions and Table A6 their quantiles. Table A4 sizes the two channels in two ways, and the allocation is less stable than the gap itself.

Different exact and sequential accounting conventions allocate the observed cost gap differently. Panel A of Table A4 substitutes historical medians into each recent episode. Replacing each recent episode's starting unemployment rate with the historical median raises the median cost from 1.16 to 1.64, closing 32 percent of the 1.49-point gap; replacing the proportional response instead raises it to 1.91, closing 51 percent. The two substitutions do not compose: replacing both reaches 2.28, which leaves 25 percent of the gap to the sizes of the declines and to how the inputs combine across episodes. Averaged over the two orders, starting unemployment accounts for 28 percent and the proportional response for 47 percent. Panel B changes the target. Taking logs, $\ln C = \ln u_0 + \ln s + \ln g(sD)$ with $g(x) = (e^x - 1)/x$, so the difference in mean log cost splits exactly into three terms, computed over the 34 historical episodes in which unemployment rose. Starting unemployment accounts for 23 percent of that difference, the proportional response for 69 percent, and the third term, a nonlinear finite-change adjustment, for 8 percent. The components characterize the observed changes; they are jointly determined within each episode. Weighting economies equally rather than episodes gives 40 percent for starting unemployment in the sequential scheme and 27 in the log scheme. Excluding the 2007–08 episodes, the historical median share equals the recent one, 0.29, the gap narrows to 2.28 against 1.16, and starting unemployment accounts for 39 percent of the smaller gap.

Table A4. Accounting for the difference in the unemployment cost of cooling

	All economies	Stock-vacancy economies
A. Sequential substitution of historical medians		
Historical median cost, H (pp per 100 log points)	2.65	2.70
Recent median cost, R	1.16	1.17
Recent with historical median s, R_s	1.91	1.88
Recent with historical median u_0, R_u	1.64	1.65
Recent with both, R_{us}	2.28	2.27
Gap, $H - R$	1.49	1.53
Share of gap closed by u_0, replaced first / second (%)	32 / 25	31 / 25
Share of gap closed by s, replaced first / second (%)	51 / 43	47 / 41
Order-averaged shares, u_0 / s / residual (%)	28 / 47 / 25	28 / 44 / 28
B. Exact decomposition of the difference in mean log cost		
Historical geometric-mean cost	2.23	2.27
Recent geometric-mean cost	1.22	1.22
Difference in mean log cost	0.606	0.616
Contribution of $\ln u_0$	0.141	0.200
Contribution of $\ln s$	0.420	0.372
Contribution of $\ln g(sD)$	0.046	0.044
Shares, u_0 / s / g (%)	23 / 69 / 8	32 / 60 / 7
Episodes with $s > 0$, historical / recent	34 / 14	29 / 13

Notes: For each episode, the unemployment cost is $C = u_0(e^{sD} - 1)/D$ (equation 1), where u_0 is the starting unemployment rate, s the unemployment share of the decline in log tightness (the proportional response), and D the decline in natural-log units. Panel A replaces s, u_0, or both in each recent episode with the historical median, keeping the episode's own D, and reports the resulting median cost in percentage points per 100 log points. Shares are the fractions of the gap $H - R$ closed by each substitution when it is made first or second; order-averaged shares average the two, and the residual is the part of the gap not closed by either substitution alone. Panel B splits the difference in mean log cost exactly into the three terms of $\ln C = \ln u_0 + \ln s + \ln g(sD)$, with $g(x) = (e^x - 1)/x$; the contributions sum to the difference up to rounding. The third term is a nonlinear finite-change adjustment that depends on both s and D. Panel B uses episodes in which unemployment rose ($s > 0$). The stock-vacancy sample excludes France and Japan, whose vacancy series measure flows of new vacancies. Source: authors' calculations.

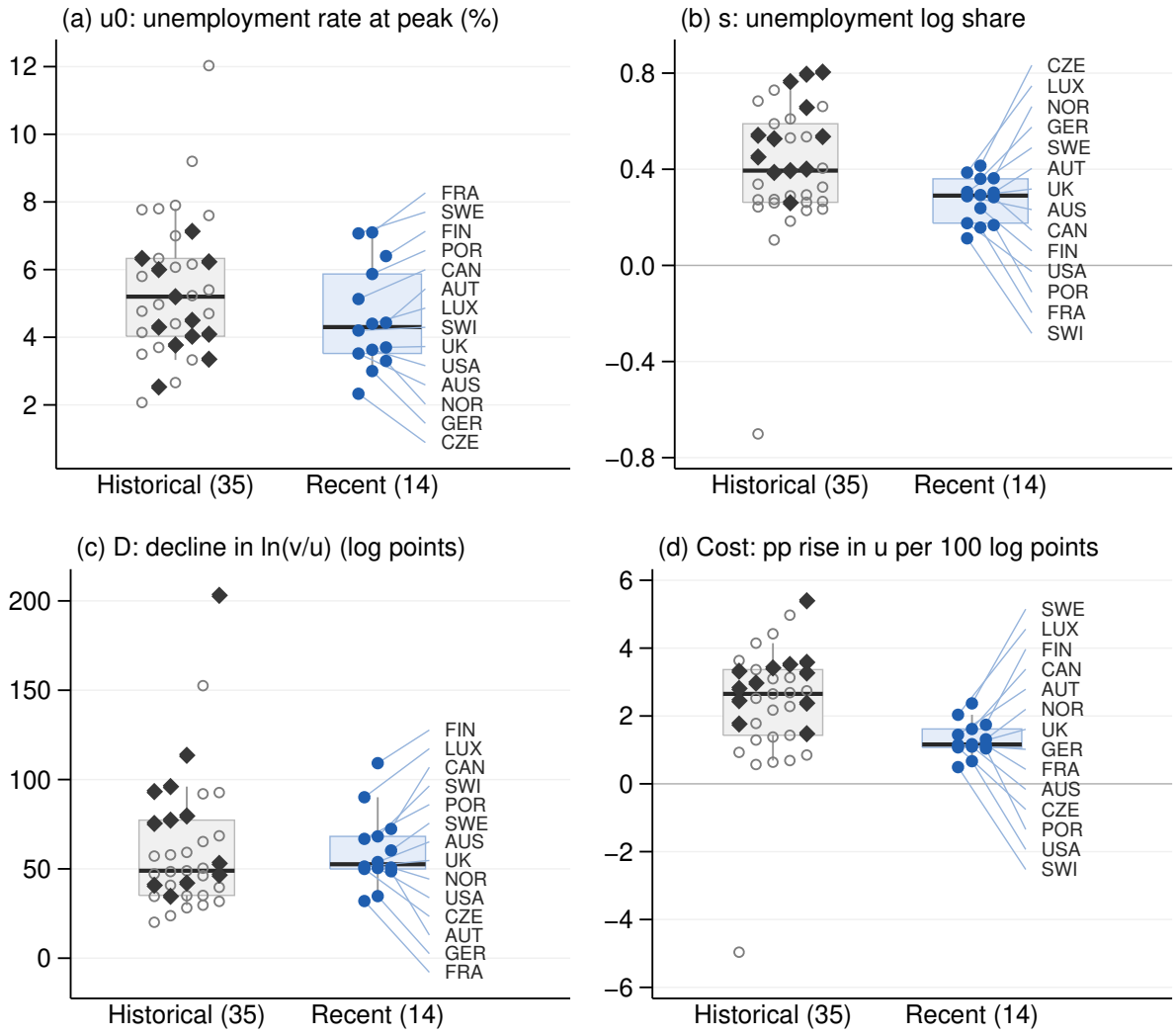


Figure A3. Distributions of the episode characteristics. Gray hollow circles are historical episodes, dark diamonds the 2007–08 episodes, and blue dots the 2022–23 episodes, labeled by economy. Boxes span the 25th to 75th percentiles, the line is the median, and whiskers span the 10th to 90th percentiles. Points are spread horizontally for legibility. Source: authors' calculations.

C Inference with dependent episodes

Uncertainty must reflect repeated observations of the same economies and a small number of global cooling waves. The 35 historical and 14 recent episodes are not independent draws: economies repeat, the recent episodes belong to one global wave, and the historical ones cluster in waves of their own. Table A5 reports the gap under three designs that each respect one form of dependence. Resampling the 17 economies with replacement puts a 95 percent interval of $[-1.87, -0.88]$ points on the gap in median cost and $[-0.85, -0.39]$ on the gap in mean log cost; the gap in the median proportional response, -0.10, has an interval of $[-0.22, 0.02]$ that includes zero. Grouping the historical peaks into 7 five-year calendar blocks and resampling blocks, with the recent episodes held fixed, gives $[-1.89, -0.62]$ for the cost gap and $[-0.24, 0.02]$ for the share gap. Treating the recent episodes as fixed, their median cost sits at the 17th percentile of the historical distribution, with 6 of 35 historical episodes below it. The shares of the gap in Table A5 have wide intervals; only the proportional response's share in the log decomposition excludes zero under both designs.

Within-economy comparisons remove differences in the composition of countries without removing common international shocks. Table A7 compares each recent episode with the same economy's historical episodes: the recent cost is lower in 14 of 14 economies, by 1.19 points in the median economy; the proportional response is lower in 12 of 14. Excluding the 2007–08 episodes from each economy's own history, 11 economies retain a comparison and the recent cost is lower in 9 of them (Figure A4). The paired tests in the table treat the economies as independent, which their common global shock violates. Table A8 lists the global cooling waves: grouping historical peaks within six quarters of one another gives 7 waves, whose median costs range from 1.4 to 4.1; the recent wave's median of 1.16 is below all of them, and resampling the 7 historical waves gives $[-2.04, -0.62]$ for the cost gap and $[-0.24, 0.02]$ for the share gap. With seven historical waves, no design can make the share difference precise, and the rank of the recent wave has a one-sided probability of $1/8$ under exchangeability.

Table A5. Inference for the recent–historical gap under three designs

		Bootstrap 95% interval		Exceedance
Statistic	Estimate	Economy	Calendar block	percentile
A. Differences, recent minus historical				
Median cost (pp per 100 log points)	−1.49	[−1.87, −0.88]	[−1.89, −0.62]	17
Median unemployment share, s	−0.10	[−0.22, 0.02]	[−0.24, 0.02]	37
Mean log cost	−0.61	[−0.85, −0.39]	[−0.82, −0.34]	15
B. Shares of the gap (%)				
Order-averaged sequential: u_0	28	[−7, 62]	[8, 65]	
Order-averaged sequential: s	47	[−2, 89]	[0, 91]	
Exact log: u_0	23	[−10, 62]	[4, 52]	
Exact log: s	69	[31, 102]	[45, 87]	

Notes: The sample contains 35 historical and 14 recent episodes. The economy bootstrap resamples the 17 economies with replacement. The calendar-block bootstrap groups historical episodes into 7 five-year blocks by peak year (1985–89, . . . , 2015–18), resamples the blocks with replacement, and holds the recent episodes fixed. Both use 999 replications (seed 1) and report percentile intervals. The exceedance percentile is the percent of historical episodes whose cost (or s) lies below the recent median, with recent episodes treated as fixed; for the mean log cost, it is the percent of the 34 historical episodes with $s > 0$ whose log cost lies below the recent mean. Panel B reports the shares of the gap in median cost (sequential substitution, averaged over the two orders) and in mean log cost (exact decomposition, episodes with $s > 0$) attributed to each input; see Table A4. Source: authors' calculations.

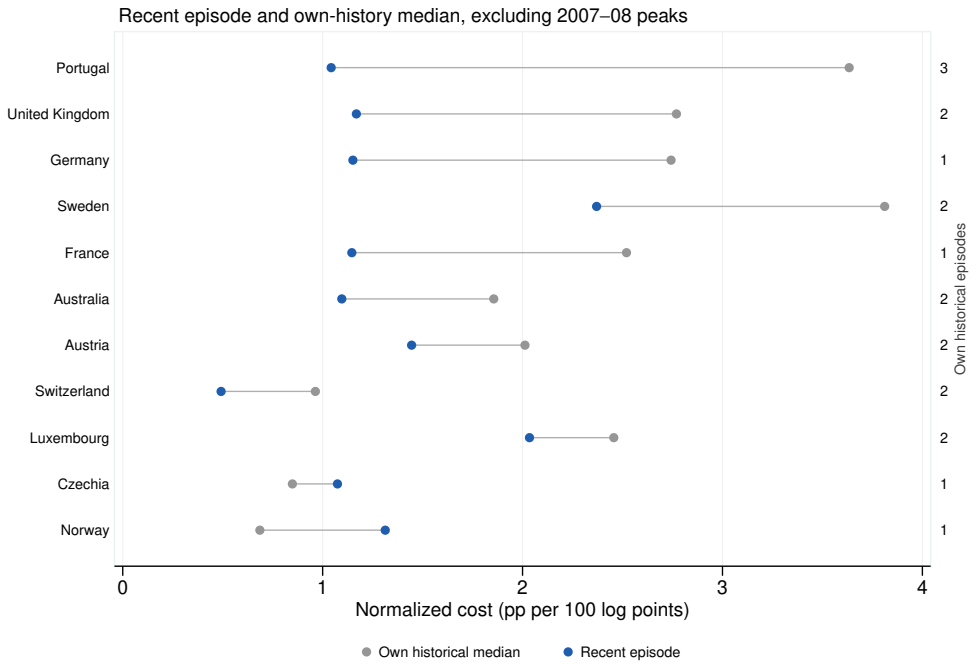


Figure A4. Recent episodes against each economy's own historical episodes outside 2007–08. Gray markers are the median cost of the economy's historical episodes with peaks outside 2007:Q1–2008:Q4, blue markers the recent episode; economies without such a comparison are omitted. Source: authors' calculations.

Table A6. Distributions of episode characteristics, historical and recent episodes

	N	P10	P25	P50	P75	P90	Mean
Starting unemployment rate, u_0 (percent)							
Historical	35	3.33	4.03	5.20	6.33	7.80	5.43
of which 2007–08	12	3.35	3.90	4.40	6.12	6.33	4.79
Recent	14	3.00	3.52	4.30	5.87	7.07	4.58
Unemployment share of the decline, s							
Historical	35	0.23	0.26	0.39	0.59	0.73	0.40
of which 2007–08	12	0.39	0.40	0.53	0.71	0.80	0.54
Recent	14	0.16	0.18	0.29	0.36	0.39	0.27
Decline in log tightness, D (log points)							
Historical	35	29.7	35.1	49.0	77.3	96.1	61.7
of which 2007–08	12	40.8	44.3	76.4	94.7	113.6	79.6
Recent	14	34.7	50.0	52.6	68.2	90.1	59.9
Increase in unemployment rate, $\Delta_8 u$ (percentage points)							
Historical	35	0.27	0.80	1.30	2.27	3.00	1.47
of which 2007–08	12	1.24	1.31	2.45	2.79	3.00	2.19
Recent	14	0.34	0.40	0.63	1.17	1.83	0.83
Unemployment cost, C (percentage points per 100 log points)							
Historical	35	0.69	1.43	2.65	3.37	4.15	2.36
of which 2007–08	12	1.76	2.41	3.12	3.47	3.58	3.03
Recent	14	0.67	1.07	1.16	1.62	2.03	1.31

Notes: The sample contains 35 historical episodes (peaks 1989–2018, of which 12 have peaks in 2007:Q1–2008:Q4) and 14 recent episodes (peaks 2021:Q3–2023:Q2). The unemployment cost C and the unemployment share s are defined in equation (1); s equals one minus the vacancy share of the decline. P10–P90 are percentiles, computed with Stata's `summarize, detail` definition. Figure A3 plots the same distributions. Source: authors' calculations.

Table A7. Recent episodes against each economy's own historical episodes

		Cost, C			Share, s		
Economy	N_{hist}	History	Recent	Diff.	History	Recent	Diff.
Australia	3	2.28	1.10	−1.19	0.29	0.29	−0.01
Austria	3	2.45	1.45	−1.00	0.53	0.30	−0.22
Canada	1	2.81	1.62	−1.19	0.39	0.28	−0.10
Czechia	2	1.16	1.07	−0.09	0.18	0.41	0.23
Finland	1	3.42	1.74	−1.68	0.45	0.24	−0.21
France	2	3.96	1.15	−2.81	0.46	0.16	−0.30
Germany	1	2.74	1.15	−1.59	0.33	0.36	0.03
Luxembourg	2	2.46	2.03	−0.42	0.44	0.39	−0.05
Norway	2	1.53	1.31	−0.22	0.49	0.36	−0.13
Portugal	3	3.63	1.04	−2.59	0.27	0.17	−0.10
Sweden	3	2.97	2.37	−0.60	0.40	0.31	−0.10
Switzerland	3	1.29	0.49	−0.80	0.29	0.11	−0.18
United Kingdom	3	3.37	1.17	−2.20	0.54	0.29	−0.25
United States	1	3.32	0.67	−2.65	0.54	0.18	−0.36
Median difference				−1.19			−0.12
Recent below history				14/14			12/14
Sign test, p				<0.01			0.01
Signed-rank test, p				<0.01			0.01

Notes: The table covers the 14 economies with a recent episode, each of which has at least one historical episode. "History" is the median over the economy's own historical episodes, or the single episode where $N_{\text{hist}} = 1$; "Diff." is recent minus history. C is the unemployment cost, in percentage points per 100 log points of declining tightness, and s the unemployment share of the decline, both defined in equation (1). The sign test is a two-sided exact binomial test, and the Wilcoxon signed-rank test is exact, computed by enumerating all 2^{14} sign patterns. Both tests treat economies as independent, although they share global shocks; Table A8 reports a comparison at the level of global waves. Figure A4 repeats the cost comparison excluding historical peaks in 2007–08. Source: authors' calculations.

Table A8. Global cooling waves

				Median	
Wave	Peaks	N	Economies	C	s
1	1989:Q1–1990:Q4	4	BEL, AUS, UK, JPN	1.37	0.27
2	1997:Q1	1	JPN	2.69	0.68
3	2000:Q2–2002:Q2	7	AUT, POR, SWI, GER, BEL, CZE, SWE	1.38	0.24
4	2004:Q1–2004:Q4	2	POR, UK	3.89	0.60
5	2007:Q1–2008:Q2	12	USA, JPN, CAN, NOR, SWE, AUS, FIN, FRA, UK, AUT, CZE, SWI	3.12	0.53
6	2010:Q1–2011:Q3	7	POR, AUS, LUX, SWI, AUT, FRA, NOR	1.78	0.27
7	2018:Q1–2018:Q4	2	SWE, LUX	4.05	0.63
Recent	2022:Q1–2023:Q1	14	CZE, NOR, POR, UK, AUT, CAN, FIN, GER, LUX, USA, AUS, SWE, SWI, FRA	1.16	0.29
Median across the 7 historical waves				2.69	0.53
Recent minus median across historical waves				−1.53	−0.24
95% interval, resampling historical waves				[−2.04, −0.62]	[−0.24, 0.02]
Rank of the recent wave among all 8 (1 = lowest)				1	4
One-sided rank probability, waves exchangeable				1/8	4/8

Notes: Two historical peaks belong to the same wave if they are at most six quarters apart, with transitive closure across economies. C is the unemployment cost (percentage points per 100 log points) and s the unemployment share of the decline, both defined in equation (1). Each wave's median is taken across its episodes; the median across waves treats each wave as one observation. Economies are listed in order of their peak quarter, with abbreviations as in Figure 1. The rank probability is the recent wave's rank among the eight wave medians divided by eight. The interval resamples the 7 historical waves with replacement (999 replications) and holds the recent wave fixed. With a four-quarter grouping rule there are 8 historical waves and the recent median cost ranks second (probability 2/9); with an eight-quarter rule there are 5 waves and it ranks first (probability 1/6). Source: authors' calculations.

D Additional benchmark results

Table A9 compares benchmark specifications and fit measures, Table A10 reports the benchmark economy by economy, and Table A11 reports four-, eight-, and twelve-quarter comparisons with bootstrap intervals. Table A12 gives the pre-2020 country elasticities behind the country-slope specification, and Table A13 reports the episodes selected on the vacancy rate alone. Figure A5 shows the residuals when crisis windows are excluded from estimation, and Figure A6 compares each recent economy's residual under the two dating rules: 7 of the 14 economies change their peak quarter, moving the median residual from 0.07 to −0.35.

Table A14 relates the benchmark residual to each economy's pre-2020 Beveridge relation, $\ln v_{it} = a_i + b_i \ln u_{it}$. Let $e_{it} = 100(\ln v_{it} - a_i - b_i \ln u_{it})$ denote the distance from the fitted relation and $r_i = 100(\Delta_8 \ln u_i + \hat{\beta}\, \Delta_8 \ln v_i)$ the benchmark residual in log points. Then

$$r_i = \hat{\beta}\, \Delta_8 e_i + (1 + \hat{\beta} b_i)\, 100\, \Delta_8 \ln u_i.$$

The recent episodes begin 51.8 log points above their fitted relations and end 19.3 above; historical episodes begin 2.5 above and end 16.7 below. Measured instead against a curve through the peak with slope $-1/\hat{\beta}$, along which the benchmark residual is zero, the recent movement is 3.6 log points, compared with 55.3 in 2007–08 and 3.3 in other historical episodes. The median fitted slope is −0.72, so the two reference curves give different distances. These measures locate observations relative to specified curves; interpreting them as changes in matching efficiency would require additional restrictions.

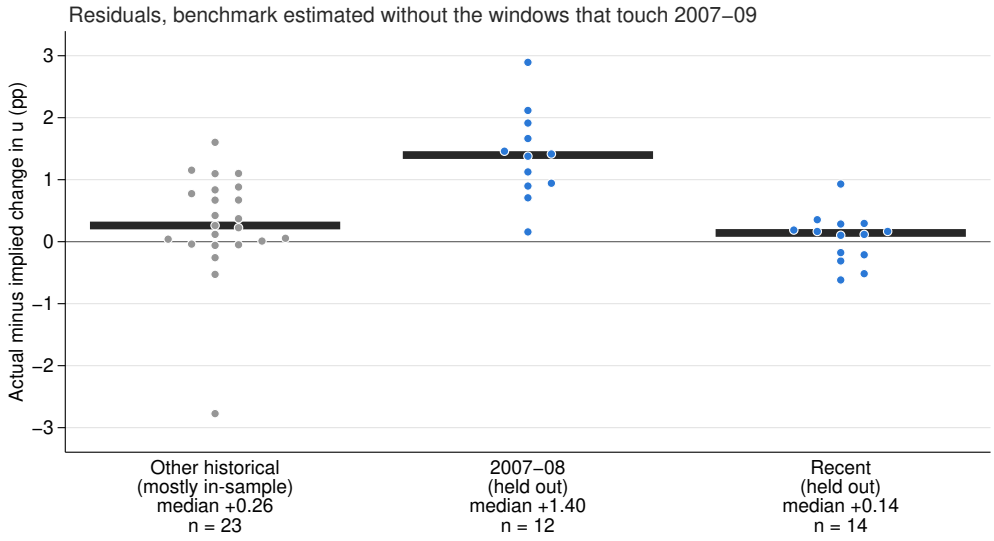


Figure A5. Benchmark residuals when the relation is estimated without windows touching 2007:Q1–2009:Q4. Actual minus implied eight-quarter change in the unemployment rate for the historical episodes outside 2007–08, the 2007–08 episodes, and the recent episodes, with group medians; the 2007–08 windows are held out of the estimation, later historical windows are not. Source: authors' calculations.

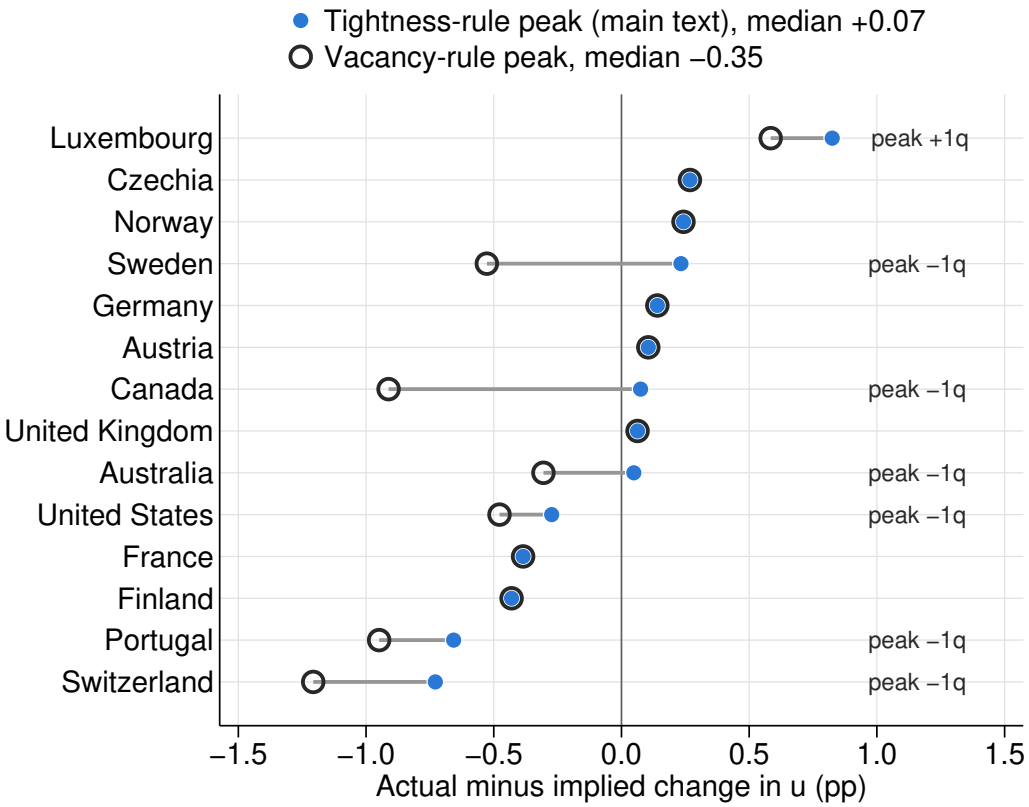


Figure A6. Benchmark residuals of the recent episodes under the two episode datings. For each economy, the residual (actual minus implied eight-quarter change in the unemployment rate, pooled pre-2020 slope) when the window starts at the tightness peak and when it starts at the vacancy peak; economies whose peak quarter differs between the rules are annotated. Source: authors' calculations.

Table A9. The recent episodes against the pre-pandemic benchmark: alternative specifications

A. Recent episodes (14; median actual increase 0.63 pp)

		Median (pp)						
Benchmark	Windows	Implied	Residual	95% CI	MAE	RMSE	Corr.	Below
Baseline	1,309	0.68	0.07	[−0.36, 0.24]	0.32	0.40	0.70	5
Economy fixed effects	1,309	0.78	−0.06	—	0.33	0.41	0.78	8
Economy-specific slopes	32–123	0.65	0.02	—	0.50	0.83	0.50	7
Leave one economy out	1,186–1,277	0.68	0.08	—	0.32	0.41	0.70	5
Excluding 2007–09	1,053	0.61	0.14	[−0.24, 0.31]	0.32	0.39	0.70	5
Excluding 2007–10	995	0.60	0.15	[−0.21, 0.33]	0.32	0.39	0.70	5

B. Historical episodes (35; peaks 1989–2018)

		Median residual (pp)			
Benchmark	Windows	Other historical	2007–08	2007–08: 95% CI	All historical
Baseline	1,309	0.06	1.27	[0.98, 1.52]	0.64
Excluding 2007–09	1,053	0.26	1.40	[1.11, 1.59]	0.71
Excluding 2007–10	995	0.35	1.42	[1.12, 1.62]	0.73

Notes: Changes are over the eight quarters after each episode's tightness peak. Every benchmark estimates equation (2) on windows in the 17 economies that end by 2019:Q4 and do not touch a documented break; implied increases follow equation (3), conditional on the realized vacancy path. The baseline uses the pooled slope with the fixed effect set to zero; "Economy fixed effects" adds each economy's estimated effect; "Economy-specific slopes" estimates the slope separately for each economy; "Leave one economy out" re-estimates the pooled slope without the economy being evaluated. Residuals are actual minus implied increases, computed for each episode before taking medians. "Windows" is the number of estimation windows; for economy-specific slopes it is the range of each recent economy's own windows, and for the leave-one-out slope it is the range across recent economies. The exclusion rows drop every window touching 2007:Q1–2009:Q4 (2007:Q1–2010:Q4). These are event holdouts rather than real-time forecasts, since later windows remain in the sample. In Panel B, the 2007–08 episodes are evaluated with a slope that never saw their windows, while most other historical windows remain in the estimation sample. Intervals are percentiles of 999 economy-cluster bootstrap replications that resample the 17 economies and re-estimate the slope. In Panel A, the interval is joint: recent episodes are resampled with their economies. In Panel B, it is for the 2007–08 median with those episodes held fixed. — indicates no bootstrap. MAE and RMSE are the mean absolute and root-mean-square residuals (pp); Corr. is the correlation of actual and implied increases across the 14 episodes; Below counts episodes with actual below implied. Source: authors' calculations.

Table A10. Recent episodes: actual and implied eight-quarter unemployment increases

				Increase in u (pp)			Residual (pp)		Percentile	
Economy	Peak	u_0	$-100\Delta_8 \ln v$	Actual	Pooled	Economy	Pooled	Economy	In-sample	LOO
A. Recent episodes										
Australia	2022:Q3	3.5	38.3	0.59	0.54	0.59	0.05	0.00	34	34
Austria	2022:Q2	4.4	33.9	0.70	0.60	0.50	0.11	0.21	37	37
Canada	2022:Q2	5.1	51.8	1.17	1.10	1.31	0.07	−0.14	37	37
Czechia	2022:Q1	2.3	29.2	0.54	0.27	0.30	0.27	0.23	37	40
Finland	2022:Q2	6.4	83.2	1.90	2.33	3.82	−0.43	−1.92	6	6
France	2023:Q1	7.1	27.0	0.37	0.75	0.78	−0.38	−0.41	6	6
Germany	2022:Q2	3.0	22.2	0.40	0.26	0.26	0.14	0.14	37	37
Luxembourg	2022:Q2	4.4	55.3	1.83	1.01	0.29	0.83	1.55	57	57
Norway	2022:Q1	3.3	32.4	0.67	0.42	0.81	0.24	−0.14	37	37
Portugal	2022:Q1	5.9	55.6	0.70	1.35	0.38	−0.66	0.32	3	3
Sweden	2022:Q3	7.1	41.9	1.43	1.20	1.38	0.23	0.05	37	37
Switzerland	2022:Q3	4.2	60.5	0.34	1.06	0.23	−0.73	0.11	3	3
United Kingdom	2022:Q1	3.7	36.3	0.60	0.54	0.70	0.06	−0.10	37	37
United States	2022:Q2	3.6	41.6	0.34	0.61	2.08	−0.27	−1.74	9	11
Median, 14 episodes		4.3	40.0	0.63	0.68	0.65	0.07	0.02	37	37
Actual below implied							5/14	7/14		
B. Economies with declines below the 20-log-point threshold										
Belgium	2022:Q1	5.3	11.2	0.30	0.23	0.13	0.07	0.18	37	37
Japan	2022:Q4	2.5	8.0	−0.03	0.08	0.11	−0.11	−0.14	20	17
Median, 16 economies		4.3	37.3	0.59	0.60	0.54	0.07	0.02	37	37
Actual below implied							6/16	8/16		

Notes: Changes are from each economy's tightness peak to eight quarters later. u_0 is the unemployment rate at the peak (percent), and $-100\Delta_8 \ln v$ is the realized decline in the log vacancy rate (log points). Implied increases follow equation (3), conditional on the realized vacancy path. "Pooled" uses the pooled slope from equation (2) ($\hat{\beta} = 0.37$) with the fixed effect set to zero; "Economy" uses each economy's own slope (Table A12). Residuals are actual minus implied increases; median rows report the median of each column, so the median residual is not the difference between the medians of actual and implied increases. Percentiles give the percent of the 35 historical episodes (peaks 1989–2018) whose pooled residual lies below the economy's. "In-sample" uses the baseline benchmark; "LOO" recomputes each historical residual from the benchmark estimated without that episode's economy. In the median rows, the percentile columns locate the median residual. Panel B lists the two economies whose tightness declined by less than the episode threshold; Spain, whose tightness rose, is omitted. Source: authors' calculations.

Table A11. Recent episodes against the pre-pandemic benchmark at four, eight, and twelve quarters

	A. 14 recent episodes			B. 16 economies		
Quarters after the peak, h	4	8	12	4	8	12
Pooled slope, $\hat{\beta}_h$	0.33	0.37	0.40	0.33	0.37	0.40
Driscoll–Kraay SE	(0.04)	(0.02)	(0.02)	(0.04)	(0.02)	(0.02)
Vacancy decline (lp)	19.0	40.0	47.5	17.0	37.3	45.0
Actual increase in u (pp)	0.30	0.63	0.84	0.24	0.59	0.79
Pooled slope						
Implied increase (pp)	0.24	0.68	0.97	0.23	0.60	0.90
90% CI	[0.18, 0.29]	[0.54, 0.83]	[0.79, 1.17]	[0.17, 0.28]	[0.48, 0.73]	[0.73, 1.09]
Residual (pp)	0.00	0.07	0.11	0.00	0.07	0.11
90% CI	[−0.04, 0.06]	[−0.06, 0.18]	[−0.07, 0.32]	[−0.04, 0.06]	[−0.06, 0.16]	[−0.07, 0.32]
95% CI	[−0.05, 0.09]	[−0.10, 0.20]	[−0.14, 0.34]	[−0.05, 0.09]	[−0.10, 0.19]	[−0.10, 0.34]
95% CI, joint	[−0.20, 0.23]	[−0.36, 0.24]	[−0.24, 0.43]	[−0.16, 0.20]	[−0.27, 0.21]	[−0.16, 0.41]
Residual (lp)	0.1	1.3	1.9	0.1	1.3	1.9
95% CI	[−1.5, 1.9]	[−2.2, 4.8]	[−2.2, 6.8]	[−1.5, 1.9]	[−2.2, 4.6]	[−2.2, 6.8]
Actual below implied	7/14	5/14	5/14	8/16	6/16	6/16
Economy-specific slopes						
Implied increase (pp)	0.25	0.65	0.78	0.24	0.54	0.66
Residual (pp)	−0.08	0.02	0.08	−0.08	0.02	0.08
Actual below implied	9/14	7/14	6/14	10/16	8/16	7/16
Windows touching a break	0	0	1	0	0	1

Notes: Entries are medians across economies of changes from each economy's tightness peak to h quarters later; pp denotes percentage points and lp log points. Panel B adds Belgium and Japan, whose tightness declined by less than the episode threshold, to the 14 recent episodes. The vacancy decline is $-100\Delta_h \ln v$. At each horizon, equation (2) is estimated with h-quarter changes on windows in the 17 economies that end by 2019:Q4 and do not touch a documented break; Driscoll–Kraay standard errors use h lags. Implied increases follow equation (3) with $\hat{\beta}_h$ and the fixed effect set to zero, conditional on the realized vacancy path. Residuals are actual minus implied increases, computed for each economy before taking medians, so the median residual is not the difference between median actual and median implied increases. CI denotes a percentile interval from 999 economy-cluster bootstrap replications (seed 1) that resample the 17 economies and re-estimate $\hat{\beta}_h$, holding the recent episodes fixed; the joint interval also resamples recent episodes with their economies. "Actual below implied" counts negative residuals, whose sign is the same in percentage points and log points. At 12 quarters, Czechia's window reaches the 2025:Q1 vacancy-register change, across which the series is chain-linked (Appendix A). Source: authors' calculations.

Table A12. Economy-specific elasticities of unemployment to vacancies, pre-2020

				Window start quarter	
Economy	$\hat{\beta}_i$	SE	Windows	First	Last
United States	1.09	0.11	69	2000:Q4	2017:Q4
Norway	0.68	0.14	57	2001:Q3	2017:Q4
Finland	0.56	0.08	80	1998:Q1	2017:Q4
Japan	0.52	0.10	123	1985:Q1	2017:Q4
United Kingdom	0.48	0.15	123	1985:Q1	2017:Q4
Canada	0.44	0.07	56	2004:Q1	2017:Q4
Sweden	0.43	0.04	68	2001:Q1	2017:Q4
Czechia	0.42	0.04	80	1998:Q1	2017:Q4
Australia	0.40	0.08	117	1985:Q1	2017:Q4
France	0.38	0.07	60	2003:Q1	2017:Q4
Germany	0.38	0.10	99	1991:Q1	2017:Q4
Austria	0.31	0.05	100	1993:Q1	2017:Q4
Spain	0.24	0.21	40	1999:Q1	2017:Q4
Belgium	0.21	0.13	81	1985:Q1	2017:Q4
Luxembourg	0.11	0.04	44	2007:Q1	2017:Q4
Portugal	0.11	0.10	80	1998:Q1	2017:Q4
Switzerland	0.09	0.02	32	2010:Q1	2017:Q4
Median across economies	0.40				
Pooled, economy fixed effects	0.37	0.04	1,309	1985:Q1	2017:Q4

Notes: $\hat{\beta}_i$ is minus the OLS coefficient from regressing the eight-quarter change in $\ln u$ on the eight-quarter change in $\ln v$, estimated separately for each economy on windows $t, \ldots, t+8$ that end by 2019:Q4 and do not touch a documented break. Economy standard errors are Newey–West with 8 lags. "Windows" is the number of overlapping eight-quarter windows, and the start-quarter columns give the first and last t. The last row is the pooled benchmark of equation (2); its standard error is clustered by economy (the Driscoll–Kraay standard error is 0.02). Belgium, Japan, and Spain enter estimation but have no recent episode. Economies are sorted by elasticity. Source: authors' calculations.

Table A13. Episodes selected on the vacancy rate alone

	Vacancy decline ≥ 20 log points				Vacancy decline ≥ 30 log points			
	Historical				Historical			
	Other	2007–08	All	Recent	Other	2007–08	All	Recent
A. Sample								
Episodes	26	10	36	14	12	7	19	11
Economies	13	10	16	14	11	7	14	11
Episodes with u observed	25	10	35	14	12	7	19	11
Tightness decline below 20 lp	4	1	5	0	0	0	0	0
Same peak as tightness rule	7	3	10	7	5	3	8	4
Within 2 quarters of tightness peak	14	8	22	14	10	7	17	11
B. Medians								
Vacancy decline (lp)	29.8	38.6	32.7	36.1	47.6	48.4	48.0	38.6
Tightness decline, D (lp)	48.5	66.9	52.5	49.6	61.2	79.6	63.8	50.8
Starting unemployment, u_0 (%)	6.2	5.7	6.2	4.4	6.2	5.2	6.2	4.4
Increase in u (pp)	0.57	1.65	1.00	0.47	1.12	2.77	1.30	0.60
Increase in $\ln u$ (lp)	12.3	23.8	15.8	10.4	16.8	36.8	21.7	8.3
C. Median residual, actual minus implied increase (pp)								
All pre-2020	−0.13	0.45	0.14	−0.35	−0.01	0.60	0.21	−0.43
Actual below implied	15/25	2/10	17/35	8/14	6/12	1/7	7/19	7/11
Excluding 2007–09	0.01	0.58	0.22	−0.21	0.22	0.71	0.31	−0.25
Actual below implied	12/25	1/10	13/35	8/14	5/12	0/7	5/19	7/11
Windows ending by 2019:Q4	−0.14		−0.01		−0.01		0.21	
Actual below implied	14/21		16/31		5/10		6/17	
D. Episodes in which tightness fell								
Episodes	23	10	33	14	12	7	19	11
Median cost, C (pp per 100 lp)	1.62	2.55	1.78	1.15	1.77	2.57	1.96	1.17

Notes: Episodes begin at a local peak in $\ln v$, with no higher value within six quarters on either side and $\ln v$ observed four quarters before and eight after, followed by a decline in $\ln v$ of at least 20 (or 30) log points over the next eight quarters. No condition is imposed on tightness or unemployment (Section 2). Historical episodes have peaks before 2019:Q1, with 2007–08 denoting peaks in 2007:Q1–2008:Q4; recent episodes have peaks in 2021:Q3–2023:Q2. All changes are over the eight quarters after the vacancy peak; pp denotes percentage points and lp log points. Panel A compares peak dates with those selected by the tightness rule. Implied increases follow equation (3) with the fixed effect set to zero, using the pooled pre-2020 slope ($\hat{\beta} = 0.37$) or the slope estimated without windows touching 2007:Q1–2009:Q4 ($\hat{\beta} = 0.34$). The 2019:Q4 row drops historical windows that reach into 2020; every 2007–08 window ends by 2010:Q4, and no recent episode qualifies. Panel D reports the unemployment cost of equation (1) only for episodes in which tightness fell, since a vacancy-selected episode need not have a large decline in tightness. Unemployment is not observed for Austria 1990:Q2, which enters only the vacancy statistics. Source: authors' calculations.

Table A14. Movement along and shifts of pre-2020 Beveridge curves over the eight quarters after the peak

	Recent		Historical (peaks 1989–2018)		
	14 episodes	16 economies	All	2007–08	Other
A. Distance from economy curves, e_{it} (log points of v)					
At the peak, median	51.8	51.8	2.5	0.1	6.8
Above the curve	13/14	15/16	20/35	6/12	14/23
Eight quarters later, median	19.3	21.0	−16.7	−3.7	−21.2
Above the curve	12/14	14/16	11/35	5/12	6/23
Latest quarter, median	18.1	18.3	—	—	—
Above the curve	11/14	13/16	—	—	—
B. Change in distance over the eight quarters (log points)					
Economy curves, median	−27.7	−21.1	−23.6	5.0	−25.7
Inward	14/14	16/16	26/35	6/12	20/23
Pooled curve, median	−27.1	−25.6	−18.8	−4.7	−21.4
Inward	14/14	16/16	26/35	8/12	18/23
Curve with slope $-1/\hat{\beta}$, median	3.6	3.4	28.9	55.3	3.3
Inward	5/14	6/16	10/35	1/12	9/23
C. Decomposition of the benchmark residual r_i (log points)					
Residual r_i, median	1.3	1.3	10.8	20.6	1.2
Shift term $\hat{\beta}\,\Delta_8 e_i$, median	−10.3	−7.9	−8.8	1.9	−9.6
Movement term, median	9.0	8.6	17.5	24.1	11.8
Corr. of r_i with $\Delta_8 e_i$, economy curves	−0.02	−0.04	0.70	0.70	0.63
Corr. of r_i with $\Delta_8 e_i$, pooled curve	0.70	0.57	0.70	0.82	0.75

Notes: Pre-2020 Beveridge curves $\ln v_{it} = a_i + b_i \ln u_{it}$ are estimated by OLS on quarterly data through 2019:Q4. Economy slopes have a median of −0.72 and are negative in 14 of 17 economies; the pooled curve has a common slope of −0.77 (Driscoll–Kraay SE 0.12) with economy intercepts. The distance $e_{it} = 100(\ln v_{it} - a_i - b_i \ln u_{it})$ is positive when an observation lies above its curve, and a negative change $\Delta_8 e_i$ is an inward movement: fewer vacancies at the realized unemployment rate. The curve with slope $-1/\hat{\beta} = -2.68$ passes through the peak, and the benchmark residual is zero along it; the change in distance from that curve equals $r_i/\hat{\beta}$. Panel C uses the identity $r_i = \hat{\beta}\,\Delta_8 e_i + (1 + \hat{\beta} b_i)\,100\,\Delta_8 \ln u_i$, where $r_i = 100(\Delta_8 \ln u_i + \hat{\beta}\,\Delta_8 \ln v_i)$ and $\hat{\beta} = 0.37$; the movement term is $(1 + \hat{\beta} b_i)\,100\,\Delta_8 \ln u_i$. The identity holds exactly for every episode and every slope b_i, but medians of the terms need not sum to the median residual. The 16 economies add Belgium and Japan to the 14 recent episodes. Historical curves include the historical observations, and 29 of the 35 historical episode windows enter benchmark estimation. Counts are episodes with the stated sign, out of those with data. Source: authors' calculations.

E Unemployment inflows and outflows

Unemployment-duration measures distinguish aggregate entry and exit hazards, but not the destinations of people leaving unemployment. Following Shimer (2012) and Elsby et al. (2013), we compute quarterly entry and exit rates from unemployment by duration for every economy with the data: Eurostat's labor-force survey for the European economies and national surveys for the United States, Canada, Australia, and Japan. In this appendix, s_t denotes the entry hazard and f_t the exit hazard; s_t is distinct from the unemployment share s in equation (1). Exits include movements into inactivity, and entries include movements from outside the labor force, so the two-state method does not separate job finding from participation changes. The baseline uses the unemployed with durations under three months; the recent U.K. episode uses a six-month threshold over a two-quarter interval. Coverage is 24 of the 35 historical episodes, 11 of the 12 from 2007–08, and all 14 recent ones. For each episode we split the change in the flow steady-state log unemployment rate, $\ln u_t^* = \ln s_t - \ln(s_t + f_t)$, exactly into entry and exit contributions over the eight-quarter window. Because u^* moves more than u over eight quarters, Table A15 also reports a dynamic decomposition of the actual change in $\ln u$.

Inferred unemployment exits deteriorated less during recent cooling under the baseline quarterly duration measure, although the inference depends on the measurement window. Figure A7 gives the result. The entry contribution in the median recent episode, 12.0 log points, is close to the historical median of 13.9. The exit contribution is where the episodes differ: exit rates fell far less than in earlier episodes, contributing 11.1 log points in the median recent episode against 20.0 in history and 24.1 in 2007–08. Given each episode's vacancy decline, the recent exit contributions sit a median 10.1 log points below the historical relation, in 11 of 14 episodes (sign test $p = 0.06$); in a pooled regression the recent shift is -6.3 log points with a standard error of 5.5, and -11.2 with a standard error of 3.6 when the rates are averaged over four quarters. The result depends on the duration window: it holds with four-quarter averages, a six-month window, and the dynamic decomposition; with a twelve-month window the recent exit contribution, 9.5, is still below the historical 15.0 but the entry contribution is too; and with the OECD's annual duration data the difference moves to entries (Table A16). Switzerland's exit contribution of 60.4 comes from a single

Table A15. Entry and exit contributions to the unemployment increase after tightness peaks: steady-state and dynamic decompositions

A. Steady-state decomposition of $\Delta_8 \ln u^*$ *(baseline)*

	Episodes	$\Delta_8 \ln u$, panel data	$\Delta_8 \ln u^*$	Entry contrib.	Exit contrib.	Entry dominates	Approx. error
Historical	24	27.1	36.6	13.9	20.0	0.29	−6.8
of which 2007–08	11	34.9	43.0	15.0	24.1	0.09	−5.1
Recent	14	15.3	30.8	12.0	11.1	0.57	−11.6

B. Exact dynamic decomposition of $\Delta_8 \ln u$

	Episodes	$\Delta_8 \ln u$, flows data	Initial condition	Entry contrib.	Exit contrib.	Entry dominates
Historical	24	27.5	−6.5	16.4	22.3	0.29
of which 2007–08	11	33.6	−1.9	20.3	22.4	0.36
Recent	14	15.7	−4.5	16.2	7.0	0.79

Notes: Medians across episodes, in log points, over the eight quarters after each tightness peak. Historical episodes have peaks in 1989–2018 and recent episodes in 2022–23; 24 of the 35 historical episodes and all 14 recent episodes have duration data. The entry hazard s_t and exit hazard f_t are not measured worker flows: they are inferred from unemployment by duration following Shimer (2012), using quarterly hazards and treating the unemployed with durations under three months as entrants (under six months over a two-quarter interval for the recent U.K. episode). Exits include moves out of the labor force as well as into employment. Panel A: $u_t^* = s_t/(s_t + f_t)$ is the flow steady state, and the entry and exit contributions split $\Delta_8 \ln u^* = \Delta_8 \ln s - \Delta_8 \ln(s + f)$ exactly using logarithmic-mean weights. The panel-data column reports the change in log unemployment in the main data set; the approximation error is the median of $\Delta_8 \ln u - \Delta_8 \ln u^*$, with u from the same source as the flows. Panel B simulates the law of motion $u_t = (1 - e^{-ha_t})\, s_t/a_t + e^{-ha_t} u_{t-h}$, with $a_t = s_t + f_t$ and h equal to one quarter (two for the recent U.K. episode), from the unemployment rate at the peak. The initial-condition term is the change with both hazards held at their peak-quarter values, which reflects convergence toward the peak-quarter steady state; the entry and exit contributions are the Shapley (order-averaged) split of the remainder. For each episode, the three terms sum exactly to $\Delta_8 \ln u$ in the flows data, but medians need not add up. "Entry dominates" is the share of episodes in which the absolute entry contribution exceeds the absolute exit contribution. Source: Eurostat, national statistical agencies, and authors' calculations.

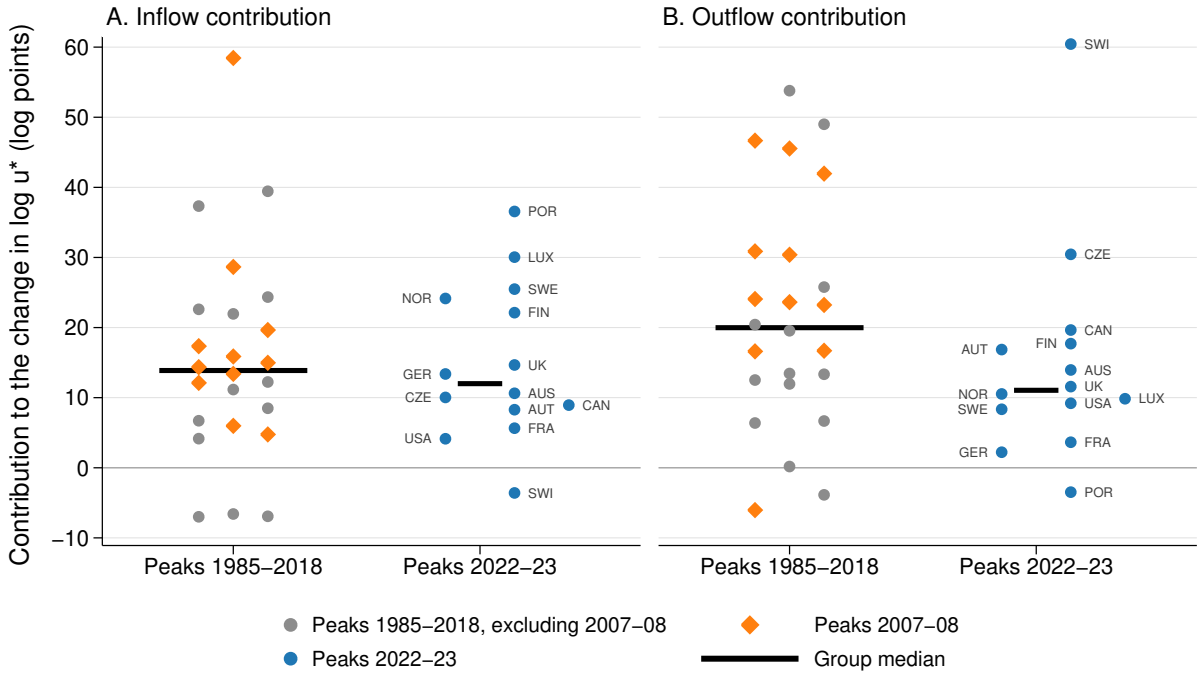


Figure A7. Contributions of entry and exit rates to the change in steady-state log unemployment over cooling episodes. Each dot is an episode; the black bar is the group median. Rates are quarterly hazards inferred from unemployment by duration under three months with the method of Shimer (2012). Source: Eurostat, national statistical agencies, and authors' calculations.

quarter, and Portugal's is negative. The evidence is consistent with several readings of the post-pandemic adjustment discussed by Barlevy et al. (2024) and Ahn and Rudd (2025) and does not distinguish among them.

Table A16. Entry and exit contributions under alternative measures

	Historical			Recent			Recent residual	
Measure	N	Entry	Exit	N	Entry	Exit	Entry	Exit
Baseline (3-month window)	24	13.9	20.0	14	12.0	11.1	−4.1	−10.1
Dynamic decomposition	24	16.4	22.3	14	16.2	7.0	1.0	−13.0
Four-quarter centered averages	24	12.7	15.3	13	9.8	4.3	−3.4	−9.9
Steady-state entry rate	24	2.7	19.9	14	1.9	11.1	−3.0	−10.0
Unknown durations allocated	24	12.5	20.2	14	12.0	11.3	−3.8	−10.1
6-month window	24	17.7	16.0	14	16.8	9.2	−4.8	−7.8
12-month window (excl. U.S.)	25	19.2	15.0	12	15.5	9.5	−5.2	−3.9
Annual OECD data (12-month)	34	16.7	10.7	14	11.1	11.9	−5.6	0.4
Excluding series breaks	17	13.4	20.4	13	13.4	11.6	−1.6	−11.3

Notes: Medians across episodes, in log points, over the eight quarters after each tightness peak. Historical episodes have peaks in 1989–2018 and recent episodes in 2022–23. Entry and exit are contributions to $\Delta_8 \ln u^*$ as in Table A15, Panel A; in the dynamic row, they are contributions to $\Delta_8 \ln u$ from simulating the law of motion, net of the initial-condition term (Panel B). The recent residual is the median, across recent episodes, of each contribution minus its value predicted by an OLS regression of that contribution on the eight-quarter decline in log vacancies across historical episodes. The steady-state entry rate sets $s_t = f_t u_t/(1-u_t)$, with u_t as a fraction, so that $u_t^* = u_t$. "Unknown durations allocated" distributes the unemployed with unknown duration pro rata. The 6- and 12-month rows change the duration threshold and continuation interval in the quarterly data; the 12-month row excludes the United States. Annual OECD data use flows in the calendar year of the peak and two years later, with a 12-month window. "Excluding series breaks" drops episodes with a Eurostat break flag within the window. Source: Eurostat, OECD, national statistical agencies, and authors' calculations.